

A successful strategy for iterated Prisoner's dilemma with any number of channels[★]

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ABSTRACT

Iterated Prisoner's Dilemma (IPD) and its variants are widely used models for studying cooperation in human societies and AI systems. This paper focuses on multichannel IPD with any finite number of channels, and examines how an agent can achieve generally high payoffs in this setting. We introduce a novel strategy that bases its decision to cooperate or defect on the difference in the cumulative number of defections between two agents. We show that this strategy naturally possesses niceness, retaliation, and forgiveness. Moreover, we theoretically analyze the performance of our proposed strategy across various scenarios, including self-play with and without errors, as well as against a diverse range of opponent strategies. Notably, we show that our strategy is invincible, ensuring that it never receives a lower expected payoff than any opponent strategy. Lastly, we show that our strategy is stable, in terms of constituting an (approximate) Nash equilibrium or subgame perfect equilibrium, under self-play conditions. Extensive simulations empirically validate the evolutionary advantages of our strategy, and showcase its potential to facilitate the emergence and sustainment of cooperation.

1. Introduction

The Prisoner's Dilemma (PD) is arguably the most well-known game in game theory. It serves as a fundamental framework for understanding how cooperation emerges and sustains by presenting players with two options: to cooperate or to defect [2,3]. Although mutual defection is the Nash equilibrium in a one-shot PD, folk theorems indicate that iterated PD (IPD) creates opportunities for cooperation to emerge and persist [4–6], raising a question that has been the focus of extensive research for decades: What constitutes a successful strategy in IPD [7–11]?

To be considered successful, a strategy should generally achieve high payoffs when competing against a range of other strategies—a criterion that emerged as central since Axelrod's tournaments [2,12]. Many strategies have been proposed with this goal in mind

[★] Parts of this work were presented in IJCAI 2024 [1]. Compared to the conference version, the theory is generalized from 2 channels to any finite number of channels, and a number of new experiments are added to empirically corroborate our theory in games with more than 2 channels.

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(see a review [13] and references therein). However, it is well-documented that no single strategy can universally dominate in terms of evolutionary stability or resist all forms of indirect invasion [14–17]. Despite this, research has identified three key principles that underpin many successful strategies: (i) be *nice*, i.e., never be the first to defect; (ii) be *retaliatory*, i.e., be able to retaliate when faced with an opponent's defection; and (iii) be *forgiving*, i.e., capable of resuming cooperation after the opponent's defection [18,19].

This paper tackles the challenge of designing successful strategies in *multichannel* IPD [20]—a notable recent development of the theory on IPD. Unlike traditional IPD where agents interact in only one game (or channel), multichannel IPD extends the traditional IPD to better represent scenarios where agents simultaneously interact across multiple games (or channels). Such situations are common in the real world, such as two colleagues working together on a project while competing for a promotion, or neighboring businesses sharing resources while competing for customers. The multichannel nature allows agents to impact strategies and outcomes in one channel through strategies and outcomes in another. A timely example is the cooperation between Microsoft and OpenAI. While these two organizations can compete directly in multiple markets, Microsoft offers the data center infrastructure and cloud services to support OpenAI's research and development; in return, OpenAI's large language models GPT-series provide Microsoft with innovative tools that can be integrated into various applications, like Copilot and Office apps. Consequently, the multichannel IPD can increase agents' bargaining power to promote overall cooperation across multiple channels. However, as one can expect, such an interdependency across multiple channels also adds more complexity to the design of a successful strategy, which exacerbates the aforementioned challenge.

We revisit two strategies that are specially designed for multichannel IPD, namely, the multichannel Win-Stay-Lose-Shift (WSLS) and Cooperate if Coordinated (CIC) strategies [20], and find that these strategies can establish full cooperation under certain conditions though, they cannot sufficiently retaliate and are vulnerable to exploitation. In other words, opponents who consistently defect can systematically take advantage of the cooperative tendencies of these strategies, leading to generally low payoffs for these strategies. Motivated by this, we propose a novel strategy, called MCSUC (MultiChannel SUCCessful). This strategy, which embodies the concept of cumulative reciprocity [21], determines an agent's choice by considering the difference in the total defection times between itself and its opponent across all channels. Depending on this cumulative difference, MCSUC adopts one of three responses: (i) full cooperation across all channels, (ii) partial cooperation in specific channels, or (iii) full defection across all channels. The rationale is that an agent should fully cooperate to foster a cooperative dynamic if the opponent has so far defected significantly less frequently than the agent, but should partially or fully defect to avoid exploitation if the opponent has defected significantly more frequently. In our preliminary analysis, we show that MCSUC inherently possesses the key properties of successful IPD strategies: niceness, retaliatory, and forgiveness.

To better understand MCSUC, we move on to theoretically analyze its performance and stability in various scenarios. We show that when both agents adopt MCSUC, full cooperation is achieved across all channels in the absence of “trembling hand” errors ([Theorem 1](#)). Even when such errors are present, the agents attain nearly optimal expected cooperation rates and payoffs ([Proposition 1](#)). Furthermore, we demonstrate that MCSUC promptly retaliates against the always defect (ALLD) strategy, effectively preventing long-term exploitation ([Theorem 2](#)), unlike existing strategies such as multichannel-WSLS and CIC. Conversely, when interacting with the always cooperate (ALLC) strategy, MCSUC resists the temptation to exploit the opponent ([Theorem 3](#)). Additionally, we show that MCSUC ensures fairness under certain conditions by achieving the same expected payoff as any opponent strategy ([Theorem 4](#)). Perhaps most interestingly, we also show that MCSUC is invincible, as its expected payoff is always equal to or greater than that of any opponent strategy, even in the presence of errors ([Theorem 5](#) and [Corollary 1](#)). It is worth emphasizing that invincibility is not the design objective of MCSUC, but rather a property that MCSUC is shown to possess. This property implies that MCSUC is never outperformed and can sometimes achieve a higher expected payoff. Last but not least, we show that under self-play conditions, MCSUC constitutes a Nash equilibrium and a subgame perfect equilibrium in the absence of errors ([Theorem 6](#)), and constitutes an approximate Nash equilibrium when the error rate is sufficiently small ([Theorem 7](#)).

Our theory looks into the use of MCSUC in a variety of scenarios; however, to address whether agents have an incentive to adopt this strategy in the first place, we turn to evolutionary experiments. We investigate the evolutionary advantage of MCSUC in two sets of evolutionary experiments: two-strategy populations and three-strategy populations. In each two-strategy population, we consider MCSUC and one of ten other strategies: multichannel-WSLS, CIC, ALLC, ALLD, Tit-for-Tat (TFT) [2], Generous Tit-for-Tat (GTFT) [7], HardMajority [22], the cumulative reciprocal strategy (CURE) [21], and both the extortion and generous strategies [8]. The population dynamics reveal that MCSUC successfully invades nine out of the ten strategies, demonstrating its evolutionary advantage over these strategies. The ALLC strategy is the only strategy that MCSUC fails to invade. However, failing to invade ALLC is a common outcome in evolutionary IPD populations [2,23], but ALLC, on the other hand, is highly susceptible to exploitation by the ALLD strategy. Motivated by this observation, we consider a three-strategy population comprising MCSUC, ALLC, and ALLD. We show that with the presence of MCSUC, the ALLD strategy is gradually eliminated, leading to a stable coexistence between MCSUC and ALLC. In other words, our strategy effectively suppresses the ALLD strategy and protects the ALLC strategy, thereby promoting and sustaining cooperation within evolutionary populations. Simulated replicator dynamics additionally suggest that our strategy restrains the ALLD strategy and protects the ALLC strategy across different error rates, R values, and mutation rates. Each of these factors significantly influences the position of the fixed points, while only the mutation rate affects their stability. More experiments that numerically validate our analytical results and examine the capability of our strategy in other multichannel games are presented in [Appendices A](#) and [D](#). In particular, [Appendix D](#) provides additional tournament results that compare MCSUC with ten representative strategies, including ALLC, ALLD, TFT, GTFT, multichannel-WSLS, CIC, HardMajority, CURE, the extortion strategy, and the generous strategy. These results show that MCSUC achieves payoffs no lower than its opponents across the tested settings. Specifically, MCSUC maintains high cooperation with ALLC, the generous strategy, and CURE, converges to defection with ALLD and the extortion strategy, and reproduces intermediate cooperation patterns when facing TFT, GTFT, multichannel-WSLS, CIC, and

Table 1

Definitions of ten multichannel strategies that MCSUC is compared against in two-strategy population settings. While multichannel-WSLS and CIC are specially developed for multichannel IPD, the other strategies, which are originally designed for traditional IPD, are extended to multichannel IPD for fair comparison.

Strategy	Definition
<i>ALLC</i>	Always cooperates in both games.
<i>ALLD</i>	Always defects in both games.
<i>TFT</i>	Chooses the same action as the opponent's previous round in each game.
<i>GTFT</i>	Chooses the same action as the opponent's previous round in each game with generosity probability that don't retaliate in the games.
<i>multichannel – WSLS</i>	Cooperate in the game k only if both agents previously took the same action in that game.
<i>CIC</i>	Fully cooperates in all channels if the agent and its opponent choose the same action in the last round, but will defect otherwise.
<i>Hard Majority</i>	Defects in each game on the first move, and keeps defecting in the game k unless the number of times the opponent has cooperated in the game k is greater than the number of times the opponent has defected in the game k .
<i>CURE</i>	Cooperate in the game k unless your opponent defects more than you and the difference exceeds your tolerance threshold in the game k .
<i>Extortion Strategy</i>	Be able to enforce an extortionate linear relation in the game k in the absence of errors.
<i>Generous Strategy</i>	Be able to enforce a generous linear relation in the game k in the absence of errors.

HardMajority. To illustrate the potential real-world implications of MCSUC, we provide two hypothetical examples that demonstrate how its multi-channel structure and cumulative reciprocity can operate in practice.¹ In addition, we provide an analysis in [Appendix I](#), examining how the number of channels m influences payoffs and cooperation.

The remainder of this paper is organized as follows. [Section 2](#) discusses related work. [Section 3](#) describes the multichannel IPD, and revisits the multichannel-WSLS and CIC strategies [20]. [Section 4](#) formally defines and theoretically analyzes our MCSUC strategy. [Section 5](#) presents empirical results in two sets of evolutionary experiments. [Section 6](#) concludes this work and discusses directions for future work.

2. Related work

Cooperation in IPD and its variants has attracted extensive interest across multiple subfields of artificial intelligence [24–26], such as multiagent systems [10,27,28], game theory [29–31], computational social science [32,33], and reinforcement learning [34–38]. In the celebrated Axelrod's tournament [2], each strategy was paired with each other strategy in IPD, and scored on the total points accumulated through the tournament. The Tit-for-Tat (TFT) strategy, which won the tournament, initially cooperates and then mirrors the opponent's previous strategy, returning cooperation for cooperation and defection for defection. After that, a number of strategies emerged, and two notable examples are Generous Tit-for-Tat (GTFT) [7] and Win-Stay-Lose-Shift (WSLS) [23]. GTFT is a more forgiving variant of TFT that occasionally cooperates even after encountering defection, while WSLS repeats its previous strategy if the resulting payoff meets a specified threshold, otherwise switching strategies—both of which were found to be able to surpass TFT in performance. Press and Dyson [8] introduced the Zero-Determinant (ZD) strategy, demonstrating that agents in IPD can unilaterally enforce a linear relationship between their own payoff and that of their opponent. A subset of ZD strategies, known as extortion strategies, enables agents to enforce a highly asymmetrical payoff ratio that favors the extorter [38]. Akin [39] showed that agents could limit an opponent's expected payoff with the “good” strategy. Wang and Lin [9] conceptualized “invincibility”, demonstrating that under error-free conditions, extortion strategies, TFT, and certain good strategies fulfill invincibility. Hao et al. [40] established frameworks that enable agents to manage both their own and their opponent's payoffs within a predefined range, thereby guiding interactions toward mutual cooperation. Li et al. [21] introduced cumulative reciprocity, which ensures fairness in terms of equal expected payoffs regardless of the opponent's strategy. We provide a summary of these strategies and evaluate their adherence to the principles of goodness, retaliation, forgiveness, and invincibility (with or without errors) in [Appendix E](#).

However, the strategies discussed above were specifically developed for traditional IPD. More recently, researchers have also investigated strategies tailored to variants of IPD [41–43]. For example, Giordano et al. [44] proposed a robust saturated strategy for multi-player variants of IPD [45]. Then, the Firm-but-Fair strategy [46] has been successful in evolutionary competitions under the alternating IPD [47]. Moreover, the ‘self-bad, partner worse’ strategy [48] stands out in asymmetric IPD [49]. In the context of multichannel IPD, Donahue et al. [20] proposed two strategies: multichannel-WSLS, which extends WSLS to allow for multiple channels, and Cooperate if Coordinated (CIC), cooperating in all channels only when the actions across all channels are the same. As we shall show in the next section, while these two strategies are able to establish full cooperation across multiple channels under certain conditions, they fail to account for cumulative past interactions and are susceptible to prolonged exploitation without an effective means of retaliation.

While the aforementioned studies typically focus on theoretical aspects, with proposed strategies amenable to formal analysis, it is worth mentioning a recent line of research that leverages deep reinforcement learning to address sequential IPD. Various mechanisms have been introduced to embed human-like motives—such as fairness [50] and social preferences [51]—into the reward structures of Markov game to encourage cooperation. Additionally, strategies like gifting [52], formal contracts [53] and opponent shaping

¹ Detailed examples are provided in [Appendix H](#).

Table 2
Payoff bi-matrix for each game.

	C	D
C	(R_k, R_k)	(S_k, T_k)
D	(T_k, S_k)	(P_k, P_k)

[36,38,54] have shown potential in fostering cooperation among agents. We refer interested readers to [34] for comprehensive reviews.

3. Background

In this section, we describe multichannel IPD, and revisit the two strategies which are specifically designed for multichannel IPD.

3.1. Multichannel IPD

Multichannel IPD [20] models the scenarios where agents interact across multiple channels, with each channel being an infinite iteration of the traditional IPD. Suppose that there are m channels, where $m \geq 2$, in a multichannel IPD (or m -channel IPD for short). For each channel $k \in \{1, 2, \dots, m\}$, the game consists of two agents X and Y , and each agent chooses to either cooperate (C) or defect (D).

When both agents choose to cooperate, they each receive a reward R . If one agent defects while the other cooperates, the defector gains the temptation payoff T , and the cooperator receives the sucker's payoff S . When both agents defect, they each receive the punishment payoff P . The payoffs for every outcome can be summarized in the following matrix Table 2:

where the payoff values must satisfy the conditions $T_k > R_k > P_k > S_k$ and $2R_k > T_k + S_k$. Under these conditions, mutual cooperation is superior to mutual defection, while defection is the dominant strategy for both agents. Therefore, the game embodies the tension between individual interest and collective interest.

The expected cooperation rate for an agent X is defined as the expected number of rounds in which an agent cooperates. Consider a focal agent X , let $a_X^k(t)$ represent X 's action in channel k at round t . The expected cooperation rate ρ_X is then given by:

$$\rho_X = \lim_{T \rightarrow \infty} \frac{1}{mT} \sum_{t=1}^T \sum_{k=1}^m I(a_X^k(t) = C), \quad (1)$$

where $I(a_X^k(t) = C)$ is an indicator function that $I(a_X^k(t) = C) = 1$ if agent X cooperates in game k at round t , and $I(a_X^k(t) = C) = 0$ otherwise. Similarly, the expected payoff for agent X in game k can be defined as:

$$r_X^k = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T r_X^k(t), \quad (2)$$

where $r_X^k(t)$ represents the payoff that agent X receives in channel k during round t . The total expected payoff for agent X across all channels is the sum of payoffs from every channel:

$$r_X = r_X^1 + r_X^2 + \dots + r_X^m. \quad (3)$$

Agent Y 's expected cooperation rate ρ_Y and expected payoff r_Y can be calculated similarly.

3.2. Multichannel-WSLS and CIC

The multichannel-WSLS strategy dictates that an agent cooperates in channel k only if both the agent and its opponent chose the same action in that channel during the previous round; otherwise, the agent defects in the next round for that channel. Similarly, the CIC strategy specifies that an agent cooperates across all channels only if both the agent and its opponent chose the same actions across all channels in the previous round. Any mismatch in actions results in defection across all channels in the following round.

Consider an agent X using either the multichannel-WSLS or CIC strategy in an m -channel IPD. Suppose that in a given round t , both X and its opponent defect. In the next round $t + 1$, X will return to cooperation regardless of whether the opponent continues to defect. This means that even if the opponent consistently defects, X will repeatedly revert to cooperation after one round of defection. As a result, X is vulnerable to exploitation by an opponent who always defects, as X fails to sustain retaliation and thus can be continuously exploited.

We further empirically evaluate the performance of these strategies when facing exploitative strategies in two-agent experiments. For each experiment, one agent adopts multichannel-WSLS or CIC, while the other agent adopts one of three exploitative strategies: the ALLD strategy, HardMajority [22], and the extortion strategy [8]. Note that the extortion strategy involves a large class of strategies; we select two of them that stand for different levels of extortion. Each experiment lasts for 1,000,000 rounds, and is repeated for 10 times. As shown in Fig. 1, under all considered settings, where the number of channels m vary in 3, 5, and 10, agents using multichannel-WSLS or CIC earned much lower average payoffs than their exploitative opponents. This confirms that

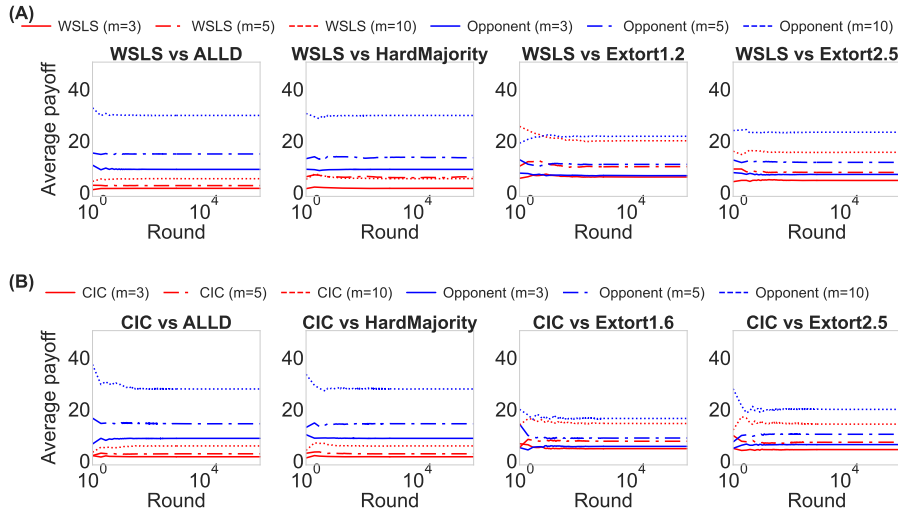


Fig. 1. The average payoffs in two-agent experiments, in which one agent (shown in blue) adopts the multichannel-WSLS or CIC strategy, while the other agent (shown in red) employs the ALLD, HardMajority, or extortion strategies. Line styles (solid, dash-dotted, and dashed) represent the results with different number of channels ($m = 3, 5$, and 10 , respectively). Under all considered settings, multichannel-WSLS or CIC tend to be exploited in the long run, and lose to the opponent strategies. The error rate $\epsilon = 1\%$. The payoff matrix is set as $R_k = 3$, $S_k = 0$, $T_k = 5$ and $P_k = 1$ for each channel k . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

these two strategies are vulnerable to long-lasting exploitation, and easily lose to exploitative strategies. In addition to the scenarios with an error rate of $\epsilon = 1\%$ shown in Fig. 1, we also conduct experiments with a lower error rate $\epsilon = 0\%$ and illustrate the results in Appendix B. The results in both cases are broadly consistent.

4. A successful strategy for multichannel IPD

In this section, we define and analyze MCSUC—our proposed strategy for multichannel IPD; the proofs are presented in Section C of the Appendix.

4.1. MCSUC

We consider that for every round, the action choice of agent X in an m -channel IPD is determined by the difference in the cumulative number of defections between two agents. Consider round t . Let $n_X^k(t)$ be the number of rounds, in which agent X chose to defect in channel k , out of the previous $t - 1$ rounds, i.e.

$$n_X^k(t) = \sum_{i=1}^{t-1} I(a_X^k(i) = D), \quad (4)$$

where $I(a_X^k(t) = D)$ is an indicator function such that $I(a_X^k(t) = D) = 1$ if agent X defects in game k at round t , otherwise, $I(a_X^k(t) = D) = 0$. For agent X , we define the cumulative number of defections to be $n_X(t) = n_X^1(t) + n_X^2(t) + \dots + n_X^m(t)$. Likewise, agent Y 's cumulative number of defections can be defined in a similar manner, and we denote this by $n_Y(t)$. Intuitively, for agent X , if its opponent Y has defected more frequently than itself, then agent X may be more prone to defection. Let $d_X(t) = n_Y(t) - n_X(t)$ represent the difference in the cumulative number of defections between two agents accumulated over the previous $t - 1$ rounds. After each round we update the cumulative defection difference $d_X(t)$ using only its value from the last round, $d_X(t - 1)$, together with the actions just taken by both agents; the next decision thus depends solely on this single value, with no need to memory earlier history. Formally, we define our strategy MCSUC for multichannel IPD as follows.

Definition 1. In an m -channel IPD, let $\Delta_{X,1}, \Delta_{X,2}, \dots$ and $\Delta_{X,m}$ be agent X 's tolerance thresholds, such that $\Delta_{X,m} \geq \Delta_{X,m-1} \geq \dots \geq \Delta_{X,1}$. At round t , agent X 's choice of actions is determined by the following rules:

- If $d_X(t) \leq \Delta_{X,1}$, agent X cooperates in all channels.
- If $\Delta_{X,k} < d_X(t) \leq \Delta_{X,k+1}$, $k \in \{1, 2, \dots, m - 1\}$, agent X defects in k channels but cooperates in $m - k$ channels. Use the priority vector $\mathbf{p}^T = [p_1, \dots, p_m]$ to determine the k channels: X defects on those with $p_i \leq k$ and cooperates on the rest. Rank channels $1, \dots, m$ in nonincreasing order of $T_k - S_k$ and set p_k to the resulting rank (smaller p_k means higher priority). Larger $T_k - S_k$ indicates a greater relative payoff from defection in channel k , hence a higher priority to defect.
- If $d_X(t) > \Delta_{X,m}$, agent X defects in all channels.

By [Definition 1](#), whenever $d_X(t)$ exceeds the next tolerance threshold, one additional channel is set to defection. The new defection is assigned to the highest-priority remaining channel under $\mathbf{p}^T$, which ranks channels in nonincreasing $T_k - S_k$ order. Larger $T_k - S_k$ means defection is relatively more attractive in channel k . Our proposed strategy MCSUC can be viewed as implementing the concept of cumulative reciprocity [21] in the context of multichannel IPD. The key idea of cumulative reciprocity is that it considers the entire history of interactions rather than just the most recent interaction. According to [Definition 1](#), if an agent employs our strategy, they will conduct a cumulative evaluation based on all past actions of the opponent and themselves to determine whether to fully cooperate, partially cooperate, or fully defect. The values of $\Delta_{X,1}, \Delta_{X,2}, \dots$ and $\Delta_{X,m}$ quantify the agent X 's tolerance in m games—the larger the value, the higher the agent's tolerance level. Priority vector $\mathbf{p}^T$ represents the agent's preference for opting to defection in different channels should the cumulative difference d_X exceeds certain thresholds. A channel with a larger value in $\mathbf{p}^T$ has a higher priority for defection. For example, considering the case where $m = 3$ and $\Delta_{X,1} = 2, \Delta_{X,2} = 4, \Delta_{X,3} = 6$, which means that this is a three-channel IPD and agent X uses MCSUC strategy with tolerance thresholds of 2, 4, 6. If $p_1 > p_2 > p_3$ and $2 < d_X^{(t)} \leq 4$, then MCSUC defects in channel 1 while cooperating in channels 2 and 3. If $p_1 > p_2 > p_3$ and $4 < d_X^{(t)} \leq 6$, MCSUC defects in channels 1 and 2 while cooperating in channel 3. If the defection difference exceeds the first threshold, the channel with the highest priority is chosen for defection; if it exceeds the second, the top two channels are chosen, and so on.

4.2. Niceness, retaliatory, and forgiveness

In traditional IPD, being nice, retaliatory, and forgiving are considered three key properties of a strategy. We extend these principles to the multichannel IPD. A strategy is nice in multichannel IPD if it never defects in any channel unless the opponent has defected first in at least one channel. A strategy is retaliatory in multichannel IPD if it responds to uncalled-for defections by defecting in one or more channels to discourage further defections and restore cooperation. Lastly, a strategy is forgiving in multichannel IPD if it is willing to let bygones be bygones and resumes cooperation in one or more channels after the opponent has defected.

Initially, MCSUC will cooperate in all games due to $d_X(t) = 0$. If the opponent defects so frequently that the difference of the cumulative defection times between the two agents exceeds MCSUC's tolerance threshold (i.e. $d_X(t) > \Delta_{X,1}$), then MCSUC will retaliate by defecting. This shows that MCSUC is nice as it is never the first to defect; on the other hand, this also indicates that MCSUC is retaliatory in response to an opponent's continuous defection. Moreover, $\Delta_{X,k}$ reflects the degree of retaliation, with a smaller $\Delta_{X,k}$ suggesting a lower likelihood of retaliating against the opponent. However, MCSUC will not retaliate endlessly. Once the opponent frequently chooses cooperation after defections, the difference in the cumulative number of defections between the two agents will decrease. Once $d_X(t) < \Delta_{X,1}$, MCSUC will revert to cooperation, demonstrating its forgiveness.

4.3. Performance under self-play

To gain some first insights into the performance, we first analyze the use of our strategy under the self-play setting, i.e., the multichannel IPD between two agents that both adopt our strategy. In the following theorem, we show that our strategy leads to the desired state of mutual cooperation in every channel and every round:

Theorem 1. *In an m -channel IPD, if agents X and Y both adopt the MCSUC strategy, will always cooperate in all channels, leading to full cooperation. Their expected cooperation rates are $\rho_X = \rho_Y = 1$, and the expected payoff are $r_X = r_Y = R_1 + R_2 + \dots + R_m$.*

Note that this theorem makes no assumption about the tolerance thresholds $\Delta_{X,1}, \dots, \Delta_{X,m}, \Delta_{Y,1}, \dots, \Delta_{Y,m}$, or the payoff values as long as satisfying the mere requirement of being a PD in each channel. Thus, under self-play, our strategy can result in mutual cooperation.

This theorem, however, has not considered the presence of errors. In reality, agents may deviate from their intended actions [55,56]. We follow prior work and introduce an error rate ϵ , where cooperation may be misimplemented as defection (and vice versa) [57], and assume $0 \leq \epsilon < 0.5$ [21,58,59]. We take the presence of errors into account in the following theorem:

Proposition 1. *In an m -channel IPD with an error rate ϵ ($0 \leq \epsilon < 0.5$), the expected cooperation rates for agents X and Y that both use MCSUC are*

$$\rho_X = \rho_Y \approx 1 - \epsilon + \frac{\epsilon - 2\epsilon^2}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1}, \quad (5)$$

which monotonically increases as the error rate ϵ decreases, or as the tolerance thresholds $\Delta_{X,1}, \Delta_{Y,1}$ increase.

As the error becomes rare, the expected cooperation rate increases; in particular, as $\epsilon \rightarrow 0$, the expected cooperation rate tends to 1, which recovers the finding in [Theorem 1](#). Moreover, if the tolerance thresholds are sufficiently large, the expected cooperation rates ρ_X, ρ_Y are approximately $1 - \epsilon$, which represents the theoretical maximum—the highest achievable cooperation rate in the presence of error. We give an analytic form of the expected payoff under this setting in Section C of the Appendix.

4.4. Performance against ALLC or ALLD

Now, we consider the scenarios in which our proposed strategy interacts with two common strategies: ALLC and ALLD. The ALLD strategy manifests an extreme case of exploitation by invariably choosing defects. By interacting with the ALLD strategy, we can gauge the capability of our proposed strategy to counteract the exploitation from the opponent. On the other hand, the ALLC strategy

represents another extreme case of always choosing cooperation. Thus, the ALLC strategy serves as a touchstone to test if our proposed strategy can resist the temptation of exploiting others.

In the following theorem, we characterize the expected cooperation rates and expected payoffs when our strategy interacts with the ALLD strategy.

Theorem 2. *In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses the ALLD strategy. With a positive error rate ϵ ($0 < \epsilon < 0.5$), both agents have the same expected cooperation rate $\rho_X = \rho_Y = \epsilon$, and have the same expected payoff*

$$r_X = r_Y = \epsilon^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + (1 - \epsilon)^2(P_1 + P_2 + \dots + P_m). \quad (6)$$

The expected cooperation rate $\rho_X = \epsilon$ implies that our strategy mostly defects and effectively acts like the ALLD strategy, when it is confronted with the ALLD strategy. Moreover, the same expected payoff as the ALLD strategy, shown by $r_X = r_Y$ in Eq. (6), suggests that our strategy can avoid lasting exploitation from the ALLD strategy, as opposed to the existing strategies (multichannel-WLS and CIC) [20].

Likewise, we obtain the following result when our strategy interacts with the ALLC strategy:

Theorem 3. *In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses the ALLC strategy. With a positive error rate ϵ ($0 < \epsilon < 0.5$), both agents have the same expected cooperation rate $\rho_X = \rho_Y = 1 - \epsilon$, and have the same expected payoff*

$$r_X = r_Y = (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m). \quad (7)$$

The expected cooperation rate $\rho_X = 1 - \epsilon$ implies that our strategy mostly cooperates, and effectively acts like the ALLC strategy, when it is confronted with the ALLC strategy. Thus, our strategy can resist the temptation to exploit the other's cooperation.

Putting Theorems 2 and 3 together, we conclude that our strategy demonstrates the characteristics of a “partner” strategy—returns cooperation for cooperation and returns defection for defection—which has been previously identified as the key to maintaining cooperation [13].

4.5. Fairness and invincibility

Based on the aforementioned results, we additionally notice an interesting finding: the expected cooperation rates of both agents are always the same, regardless of the specific scenario; this pattern also holds for the expected payoffs. This inspires us to ask whether our proposed strategy can ensure fairness in certain scenarios. We answer this question in the following theorem.

Theorem 4. *In an m -channel IPD, suppose agent X uses MCSUC while opponent agent Y uses a strategy α . For any opponent strategy α , both agents always have the same expected cooperation rate $\rho_X = \rho_Y$. Moreover, if the m -channel IPD satisfies $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$, both agents also have the same expected payoff $r_X = r_Y$.*

This theorem shows that under the condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$, our proposed strategy ensures fairness in terms of the expected payoffs, no matter what strategy the other agent plays. This condition refers to the scenarios where the difference between the temptation to defect (denoted by T) and the risk of being deceived (denoted by S) is the same across different channels.

It is then natural to ask what happens when the aforementioned condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied. Interestingly, we show in the following theorem that our proposed strategy can achieve invincibility in these scenarios. That is to say, even though fairness is not guaranteed, our strategy results in an expected payoff that is not smaller than that of any opponent strategy.

Theorem 5. *In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses a strategy α . For any opponent strategy α , if condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied, $r_X \geq r_Y$ is ensured.*

Therefore, when $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied, our strategy never loses to any opponent strategy in terms of the expected payoff.

Putting Theorems 4 and 5 together, we conclude the performance of our proposed strategy against any opponent strategy as follows:

Corollary 1. *In an m -channel IPD, suppose agent X adopts MCSUC while the opponent agent Y uses a strategy α . For any opponent strategy α , there always exists a priority vector $\mathbf{p}^T$ such that the expected payoff of our strategy is not smaller than that of the opponent strategy, i.e. $r_X \geq r_Y$.*

Simply put, our proposed strategy can always ensure invincibility, regardless of the opponent strategy. This invincibility is a property of MCSUC, not its design objective. Note that this capability of ensuring invincibility is a distinct property of our proposed strategy. In contrast, the multichannel-WLS and CIC typically fail to ensure invincibility or fairness as we have already shown in Fig. 1.

4.6. Stability

Last but not least, we examine the stability of our proposed strategy through equilibrium analysis. We focus on the self-play settings, as the equilibrium analysis given any arbitrary opponent strategy is typically intractable in IPD [8,39].

Theorem 6. *In an m -channel IPD without the presence of errors, the strategy profile $(MCSUC, MCSUC)$ is a Nash equilibrium as well as a subgame perfect equilibrium.*

This theorem indicates that in the absence of errors, if both agents use our strategy, the strategy is stable in the sense that no player can increase the payoff in the two-channel IPD as well as in every subgame by unilaterally deviating from our strategy. However, as we show in the following proposition, this does not hold given the presence of errors:

Proposition 2. *In an m -channel IPD with a positive error rate ϵ ($0 < \epsilon < 0.5$), the strategy profile $(MCSUC, MCSUC)$ is not a Nash equilibrium.*

That said, under the self-play setting, even though the use of our strategy does not lead to a Nash equilibrium, we show in the following theorem that it results in an approximate Nash equilibrium given a sufficiently small error rate:

Theorem 7. *In an m -channel IPD with a positive error rate ϵ ($0 < \epsilon < 0.5$), given a non-negative $\hat{\epsilon}$, the strategy profile $(MCSUC, MCSUC)$ is an $\hat{\epsilon}$ -equilibrium if $\hat{\epsilon}$ and for all $\epsilon \leq \frac{3\hat{\epsilon}}{m \max \left[(R_{p_1} - P_{p_1}), (2R_{p_1} - S_{p_1} - T_{p_1}) \right]}$*

For an $\hat{\epsilon}$ -equilibrium, the strategy profile approximately satisfies the condition of Nash equilibrium in the sense that no players can gain more than $\hat{\epsilon}$ in the expected payoff by unilaterally deviating from the equilibrium strategy [60]. Thus, this theorem shows that under the self-play setting, given a sufficiently small error rate, the use of our strategy is stable in the sense of satisfying the requirement of $\hat{\epsilon}$ -equilibrium.

5. Evolutionary experiments

The theory presented in the last section analyzes the use of MCSUC in a variety of scenarios, but does not take into account if agents have an incentive to adopt the strategy in the first place. To explore this latter question, in this section, we present two sets of evolutionary experiments: two-strategy populations and three-strategy populations. More experiments that numerically validate our analytical results and examine the capability of our strategy in other multichannel games are presented in [Appendices A and D](#).

5.1. Two-strategy populations

To investigate whether our strategy exhibits an evolutionary advantage in multichannel IPD, we follow the approach [21], and conduct experiments on two-strategy populations, allowing agents' strategies to evolve over time. In this setting, the population consists of an infinite number of agents, each adopting one of the two strategies. All agents are part of a well-mixed population, where every agent interacts equally with every other agent. One strategy is MCSUC ($\Delta_k = 2k$ for $k \leq m$), while the other is one of the multichannel-WSLS, CIC, ALLC, ALLD, TFT, GTFT, HardMajority, CURE, the extortion, or the generous strategies. Multichannel-WSLS and CIC are strategies specifically designed for the multichannel setting, whereas the other strategies were originally developed for traditional IPD. To ensure a fair comparison, we extend all strategies to the multichannel IPD and apply the extension in simulations. Details of these extensions are provided in shown in [Table 1](#).

Each simulation involves two steps. First, we conduct 1000 simulation runs, each lasting 1,000,000 rounds, to obtain the average payoffs of the two strategies. Second, using a noisy "survival of the fittest" environment with a mutation rate of 10% [21,61], we calculate the strategies' frequencies based on the average payoffs from the first step. When a mutation occurs, all other strategies decrease their frequencies to 99.9%, while a strategy is randomly selected to increase its frequency by 0.1%. In this noisy 'survival of the fittest' environment, we model a population where agents adopt one of the two strategies. The probability that an agent changes its strategy is largely influenced by the fitness of each strategy, determined from the average payoffs obtained earlier. Our calculation of fitness follows the approach of Nowak and Sigmund [7]. A strategy is considered capable of invasion if it starts with a low initial frequency but eventually dominates the population. A simulation concludes when the frequencies of the strategies stabilize, indicating one of two possible steady states: either the complete dominance of one strategy over the other or the stable coexistence of both strategies. It is noteworthy that, unlike the experiments shown in [Fig. 1](#)—where one agent competes against another—here we simulate a large population of agents. In this setting, some agents employing MCSUC compete against others using a different strategy.

As illustrated in [Fig. 2\(a\)](#) and (c)–(j), our strategy successfully invades nine out of the ten strategies, with the exception of the ALLC strategy. This demonstrates that our strategy possesses an evolutionary advantage over these strategies, except for ALLC. Remarkably, even when only 0.1% of agents in the population initially adopt our strategy, it completely invades the ALLD, HardMajority, TFT, GTFT, extortion, and generous strategies. Additionally, with less than 50% initial frequency, our strategy fully invades the multichannel-WSLS, CIC, and CURE strategies. We also observe that our strategy is risk-dominant compared to multichannel-WSLS, CIC, and CURE.

When competing against the ALLC strategy ([Fig. 2\(b\)](#)), our strategy does not achieve invasion. Instead, both strategies coexist stably by the end of the simulation. It is important to note that failures to invade the ALLC strategy are common in IPD research.

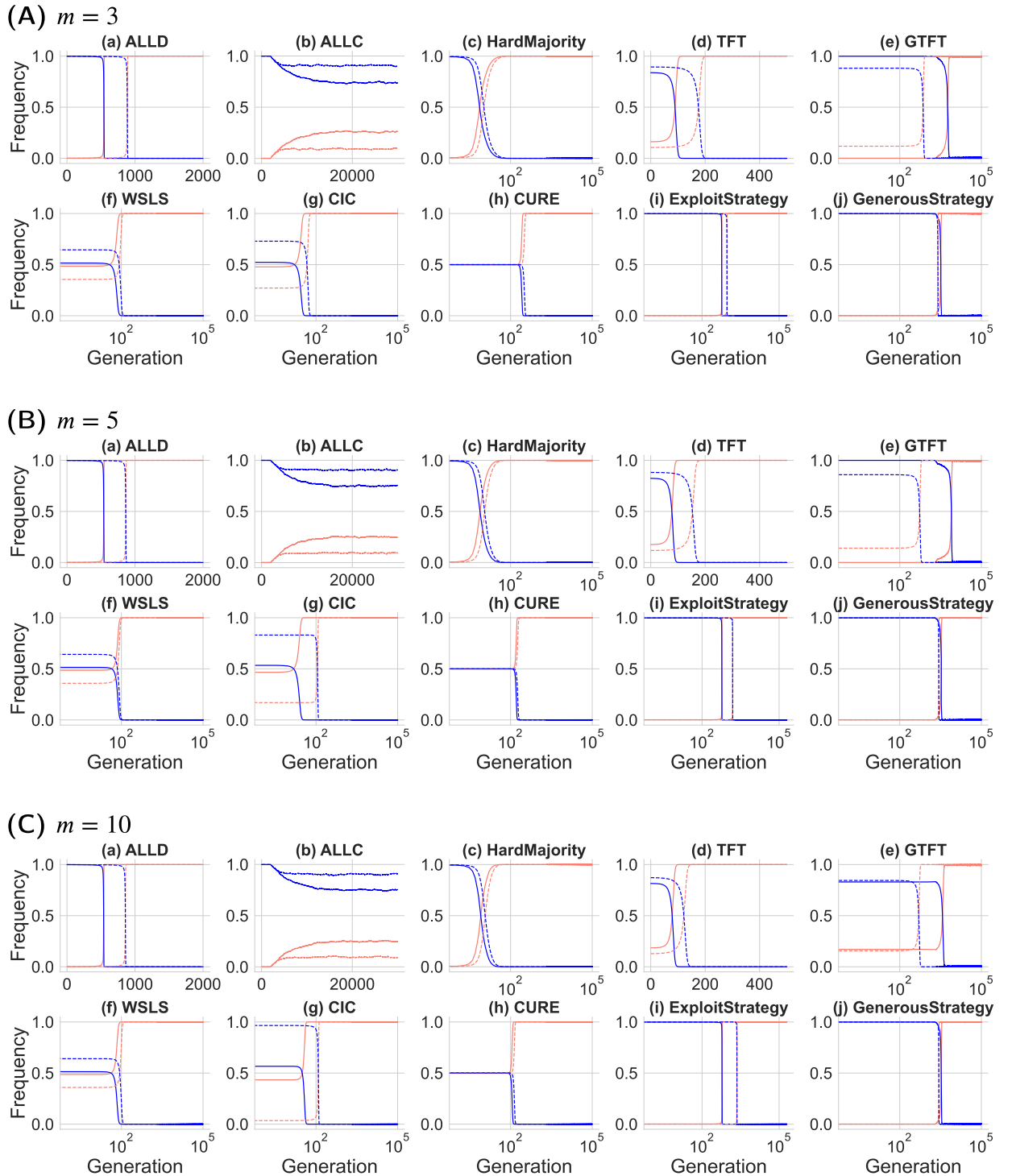


Fig. 2. The frequency of strategy use in two-strategy populations, where one strategy is MCSUC (shown in red), and the other strategy (shown in blue) is ALLD, ALLC, HardMajority, TFT, GTFT, multichannel-WSLS, CIC, CURE, the extortion strategy, or the generous strategy. Line styles (solid and dashed) represent the results with different error rates ($\epsilon = 0.01$ and 0.1 , respectively). The initial frequency of MCSUC is 0.1% in most populations and remains below 50% in others. MCSUC successfully invades nine out of the ten strategies except for ALLC. This indicates that MCSUC strategy is evolutionarily more advantaged than nine of the ten strategies, except for the ALLC strategy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

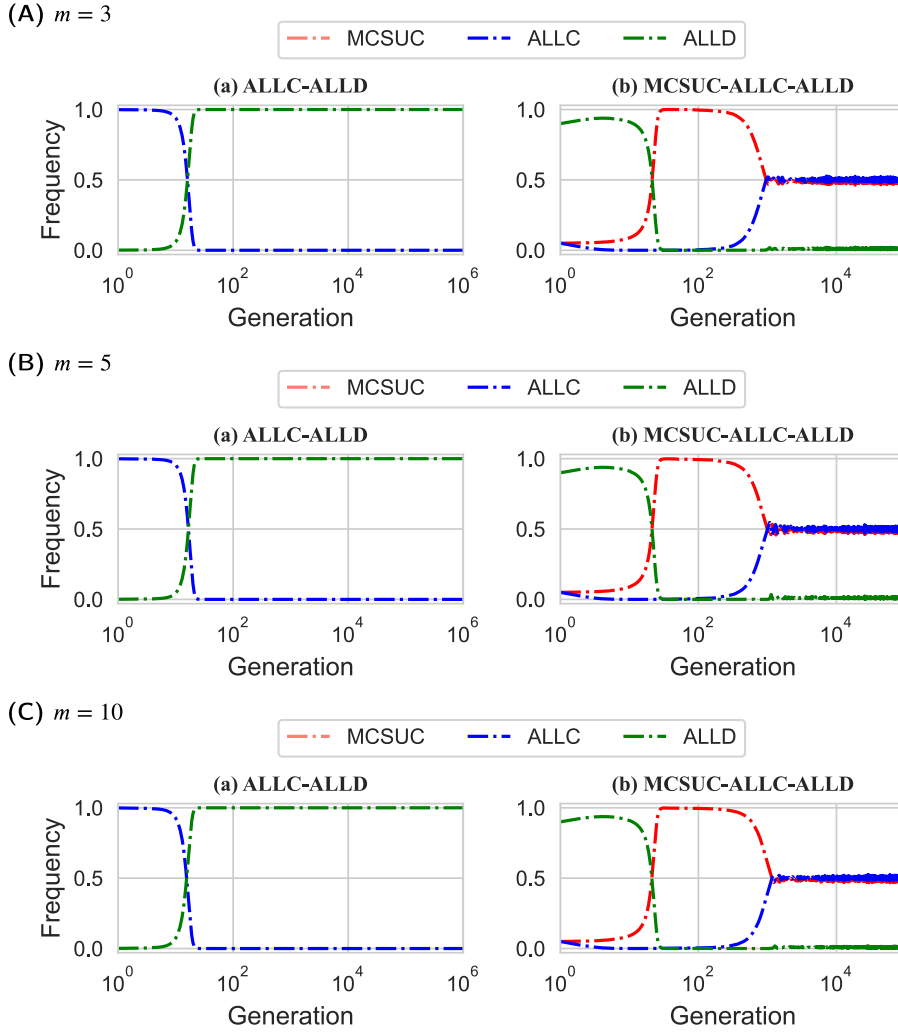


Fig. 3. The frequency of strategy use in two-strategy populations including ALLC and ALLD (left panel), and three-strategy populations including ALLC, ALLD and MCSUC (right panel) with $m = 3$ (A), $m = 5$ (B), and $m = 10$ (C). In all left panels of (A), (B), and (C), representing the two-strategy population, the ALLD strategy invades and ultimately dominates the ALLC strategy. In all right panels, representing the three-strategy population, the ALLD strategy almost disappears, and our strategy MCSUC and the ALLC strategy stably co-exist. These findings demonstrate that our strategy MCSUC effectively suppresses the ALLD strategy and protects the ALLC strategy, thereby promoting and sustaining cooperation within evolutionary populations. The error rate is fixed at $\epsilon = 10\%$. The mutation rate is set to 10% .

For instance, TFT and invincible strategies also fail to invade ALLC. Moreover, the ALLC strategy itself can be invaded by the ALLD strategy (Fig. E.5(a)), which, in turn, can be invaded by our strategy. This phenomenon encourages us to explore what will occur when ALLC, ALLD, and MCSUC coexist.

5.2. Three-strategy population

To answer the above question, we then consider a three-strategy population (with the ALLC, ALLD, and MCSUC) using a similar approach as in the two-strategy populations Fig. 3. The population consists of an infinite number of agents, each adopting one of the three strategies. All agents are part of a well-mixed population, where every agent interacts equally with every other agent. As shown in Fig. E.5, after 70 generations, the ALLD strategy is almost eliminated from the population, while most agents in the population adopt our strategy. Later, as our strategy cannot invade the ALLC strategy, the number of agents adopting the ALLC strategy increases, which eventually leads to the stable co-existence of our strategy and the ALLC strategy. Because our strategy behaves like the ALLC strategy when facing the ALLC strategy, the eventual stable co-existence suggests that most of the agents will cooperate. Therefore, although our strategy cannot invade the ALLC strategy in the two-strategy population, our strategy can protect the ALLC strategy from exploitation in the three-strategy population with the presence of the ALLD strategy, thereby promoting and sustaining cooperation.

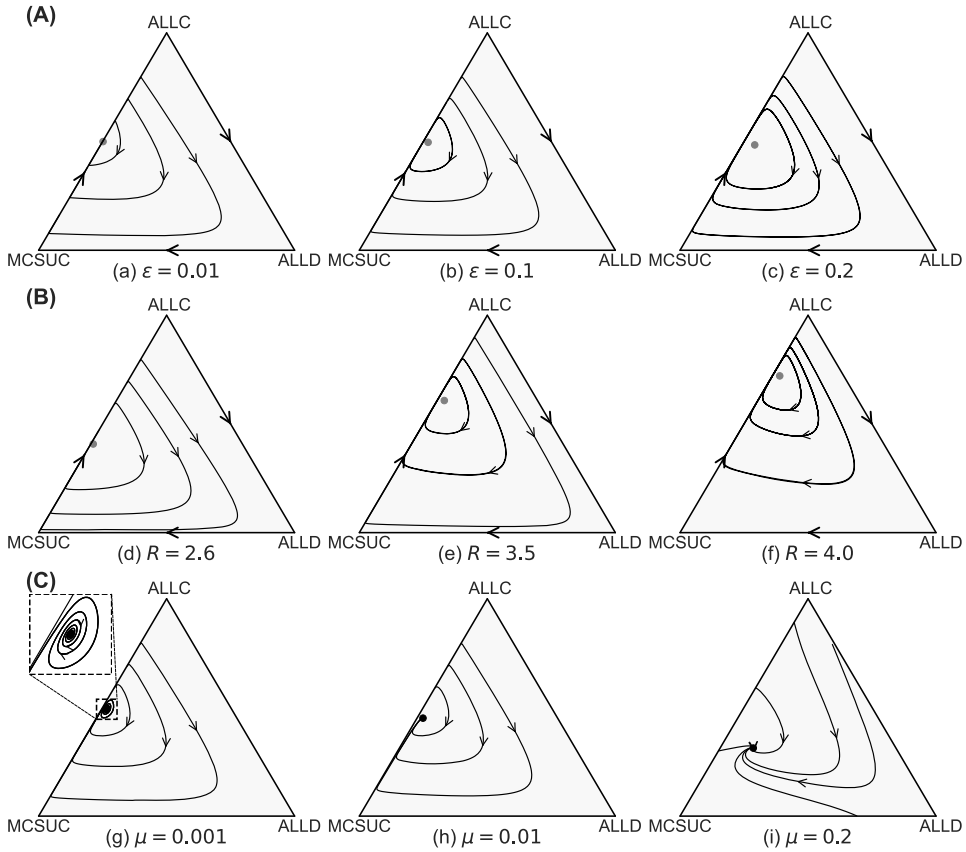


Fig. 4. Replicator dynamics for ALLC, ALLD, and MCSUC strategies, using parameters $R_k = 3$, $S_k = 0$, $T_k = 5$, $P_k = 1$ for each channel, $\Delta_{x,1} = 1$, $\epsilon = 0.1$, and $\mu = 0$, unless otherwise specified. (A), (B), and (C) illustrate replicator dynamics simulations for a three-strategy population (ALLC, ALLD, and MCSUC). Each corner represents a homogeneous population, each edge represents a population in which two of the three strategies coexist. The point represents the unique fixed point of the dynamics. Curves in the interior indicate representative trajectories. The first row (A) shows dynamics for $\epsilon = 0.01, 0.1$, and 0.2 , respectively, with the fixed point gradually shifting away from the boundary between ALLC and MCSUC as ϵ increases. The second row (B) depicts results for $R = 2.6, 3.5$, and 4.0 , respectively, where the fixed point moves along the boundary between MCSUC and ALLC, approaching ALLC as R increases, consistent with Eq. (C. 41). The third row (C) illustrates dynamics for mutation rates $\mu = 0.001, 0.1$, and 0.2 . Here, all orbits converge to the interior fixed point. As μ increases, the fixed point shifts from the edge between MCSUC and ALLC toward the center of the simplex, as described by Eq. (C. 42). This indicates that our strategy can restrain the ALLD strategy and protect the ALLC strategy across different error rates, R values, and mutation rates. Each of these factors significantly influences the position of the fixed points, while only the mutation rate affects their stability.

To further explore the evolutionary characters among these three strategies, we simulate the replicator dynamics among ALLD, ALLC, and MCSUC. As shown in Fig. 4(A) and (B), there is a rock-paper-scissors-like dominant cycle: if MCSUC is absent, ALLD dominates; if ALLC is absent, MCSUC dominates; and if ALLD is absent, ALLC dominates. Moreover, in the limit of zero error ($\epsilon \rightarrow 0$), the equilibrium is characterized by:

$$x_D^* = 0, \quad x_C^* = \frac{R_1 - P_1 + R_2 - P_2 + \dots + R_m - P_m}{T_1 - P_1 + T_2 - P_2 + \dots + T_m - P_m}, \quad x_M^* = \frac{T_1 - R_1 + T_2 - R_2 + \dots + T_m - R_m}{T_1 - P_1 + T_2 - P_2 + \dots + T_m - P_m}. \quad (8)$$

This implies that in this case, ALLD completely vanishes, and MCSUC and ALLC coexist. For positive error rates, all such interior fixed points are encircled by closed periodic orbits, indicating that nearly all initial populations lead to oscillatory dynamics among ALLC, ALLD, and MCSUC, as shown in Fig. 4(A) and (B). Furthermore, as ϵ increases from 0.01 to 0.2 (Fig. 4(A)), the fixed point moves away from populations where ALLD nearly vanishes, suggesting that higher error rates allow for the existence of the ALLD strategy. As R increases from 2.6 to 4.0 (Fig. 4(B)), the fixed point shifts from near MCSUC to near ALLC. In other words, higher values of R result in the ALLC strategy becoming more dominant relative to MCSUC.

Additionally, when mutations are possible ($\mu > 0$), we find that the oscillations around the fixed point break down. Instead, all initial populations converge to the unique fixed point, resulting in a unique and globally stable fixed point, as shown in Fig. 4(C). As μ increases, the fixed point moves from the edge between ALLC and MCSUC toward the center of the simplex, indicating that higher mutation rates facilitate a more heterogeneous coexistence among the strategies.

Overall, based on three-strategy population evolutionary experiments and replicator dynamics among ALLC, ALLD and MCSUC, we find that our strategy effectively suppresses the ALLD strategy and protects the ALLC strategy, thereby promoting and sustaining cooperation within evolutionary populations.

6. Conclusions

In this paper, we address the challenge of what a successful strategy should be in the multichannel IPD. We start with analyses of the existing strategies, showing that they are vulnerable to long-lasting exploitation and lose to some common strategies. Motivated by this, we propose a novel strategy that is nice, retaliatory, forgiving, and perhaps most interestingly, invincible. We present extensive analytical results regarding the performance and stability of our strategy under various scenarios. Furthermore, we provide empirical evidence illustrating the evolutionary advantage of our strategy against nine existing strategies through two-strategy population evolutionary experiments. Additionally, extensive simulations empirically showcase its potential to facilitate the emergence and sustainment of cooperation. Crucially, its success rests on a deliberate trade-off: it willingly forgoes part of the exploitable surplus against unconditional cooperators (e.g., ALLC) to secure long-run, sustainable cooperation with retaliatory strategies (e.g., TFT) and to reduce the risk of recurrent exploitation by persistent defectors (e.g., ALLD), thereby enhancing robustness and expected payoffs in environments with unknown, heterogeneous opponents.

As future work, several promising directions can be explored. In this paper, our focus is on proposing and analytically characterizing a successful strategy. However, whether such a strategy can be learned and adopted by boundedly rational agents in practice remains an open question. Therefore, the learnability and emergent behavior of MCSUC—particularly within multi-agent reinforcement learning frameworks, and under conditions such as bounded rationality, partial observability, or noisy interactions—warrants further investigation. Moreover, while our approach is developed in the context of the IPD, the underlying ideas are not limited to this particular game. It would be interesting to explore the generalizability of the strategy to other types of multichannel games, which may feature fundamentally different equilibrium structures. In such settings, successful strategies may exhibit new or altered properties. Investigating whether MCSUC-like strategies can adapt or evolve distinctively under different payoff structures and interaction topologies could further deepen our understanding of cooperation dynamics in complex environments.

CRedit authorship contribution statement

Zhaoheng Cao: Writing – original draft, Visualization, Validation, Methodology, Investigation; **Juan Shi:** Writing – review & editing, Investigation, Formal analysis; **Zhen Wang:** Supervision, Resources, Project administration, Funding acquisition; **Shuyue Hu:** Writing – review & editing, Visualization, Supervision, Project administration, Conceptualization; **Chen Chu:** Supervision, Project administration, Methodology, Investigation, Conceptualization.

Data availability

We will submit all the source code and data when the paper is accepted.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhen Wang reports financial support was provided by National Natural Science Foundation of China. Chen Chu reports financial support was provided by National Natural Science Foundation of China. Chen Chu reports financial support was provided by Department of Science and Technology of Yunnan Province, China. Zhen Wang reports financial support was provided by Tencent Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Robustness of our strategy

A.1. IPD with different payoff matrices

In the main paper, with the classical payoff matrix in the repeated prisoner's dilemma used by Axelrod [2], we provide empirical evidence illustrating the evolutionary advantage of our strategy against nine existing strategies through two-strategy populations.

Additionally, we demonstrate its capability to promote and sustain cooperation in evolutionary populations. Here, we are interested to see if our strategy can play a similar role in a multichannel game with different payoff matrices between games.

To ensure comparability, the payoff matrix for each channel is randomly selected from either $R = 3, S = 0, T = 5, P = 1$ or $R = 3, S = -1, T = 6, P = 1$. With the same error rate and mutation rate, it can be observed that our strategy successfully invades nine out of the ten strategies, except the ALLC strategy, again Fig. A. 1. Our results are robust in the m-channel IPD with different payoff matrices between games.

A.2. Iterated social dilemmas

In the main paper, we claim that our strategy is robust in the more general context of multichannel iterated social dilemmas. To validate the robustness of our strategy, we consider the m-channel iterated social dilemmas, such as stag hunt and snowdrift. The payoff matrix is set as $R = 3, T = 2.5, S = -0.5, P = 1$ to ensure that each channel itself is a stag hunt game. Similarly, the payoff matrix of the m-channel iterated snowdrift dilemma is set as $R = 3, T = 4.5, S = 1.5, P = 1$. We then consider two-strategy populations and let agents' strategies evolve. One strategy is MCSUC, and the other strategy is one of the multichannel-WSLS, CIC, ALLC, ALLD, TFT, GTFT, HardMajority, CURE, the extortion strategy, or the generous strategy. To ensure a fair comparison, we extend all strategies to the multichannel IPD and apply the extension in simulations. Details of these extensions are provided in shown in Table 1.

The results show that our strategy successfully invades nine of ten strategies except for the ALLC strategy, which means that our strategy is evolutionarily more advantaged than nine of the ten strategies except for the ALLC strategy. The results in the three-channel, five-channel, and ten-channel iterated stag hunt dilemma Fig. A. 2 and iterated snowdrift dilemma Fig. A. 3 are consistent with the results presented in the main paper.

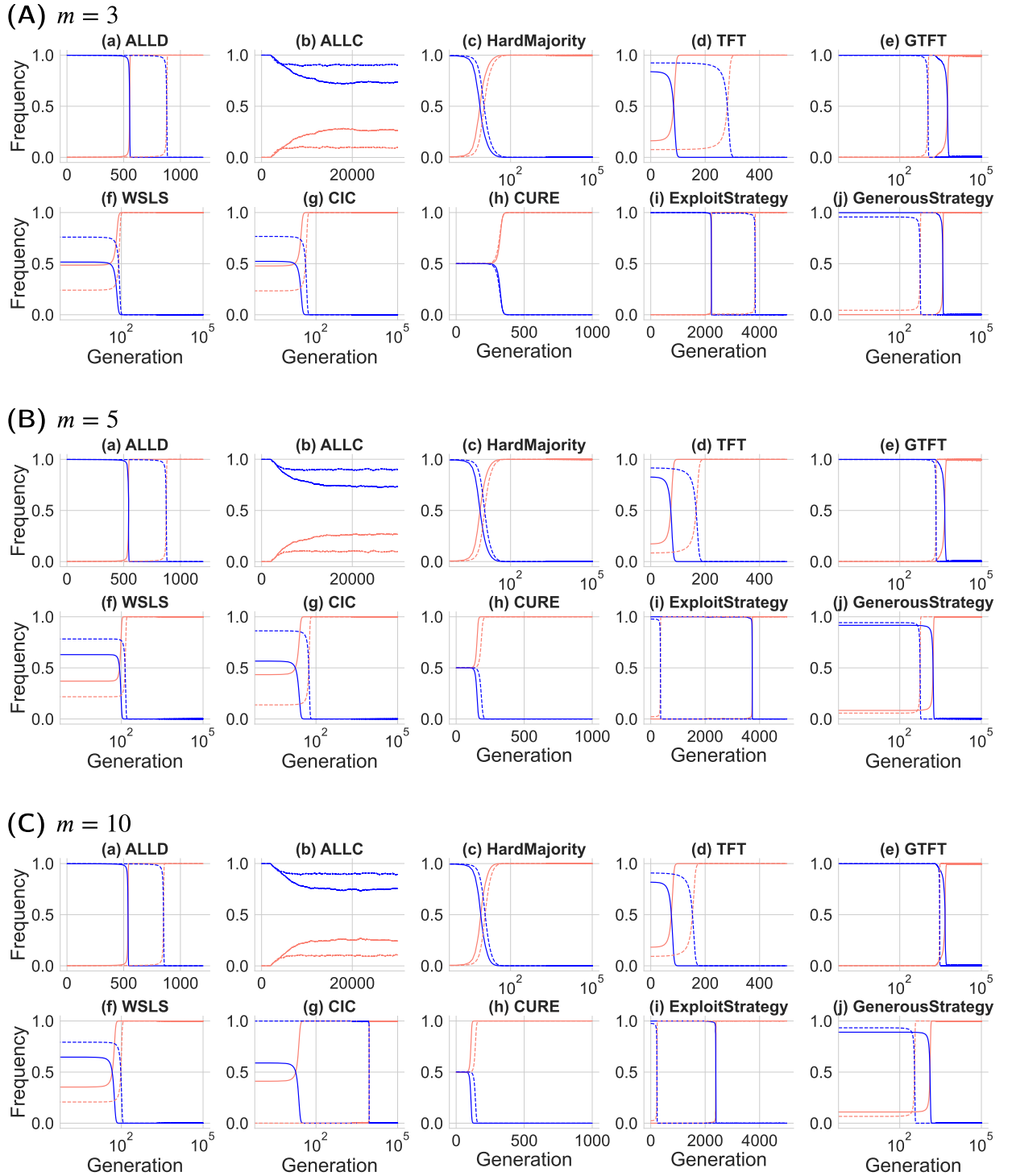


Fig. A. 1. Two-strategy populations with different payoff matrices, where one strategy is MCSUC, with $m = 3$ channels (A), $m = 5$ channels (B), and $m = 10$ channels (C). There are two error rates: $\epsilon = 1\%$ (solid line) and $\epsilon = 10\%$ (dashed line). The red line represents MCSUC, while the blue line represents the opponent's strategy. The initial frequency of MCSUC is 0.1% in most populations and remains below 50% in others. MCSUC successfully invades nine out of the ten strategies except for ALLC. This indicates that MCSUC strategy is evolutionarily more advantaged than nine of the ten strategies, except for the ALLC strategy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

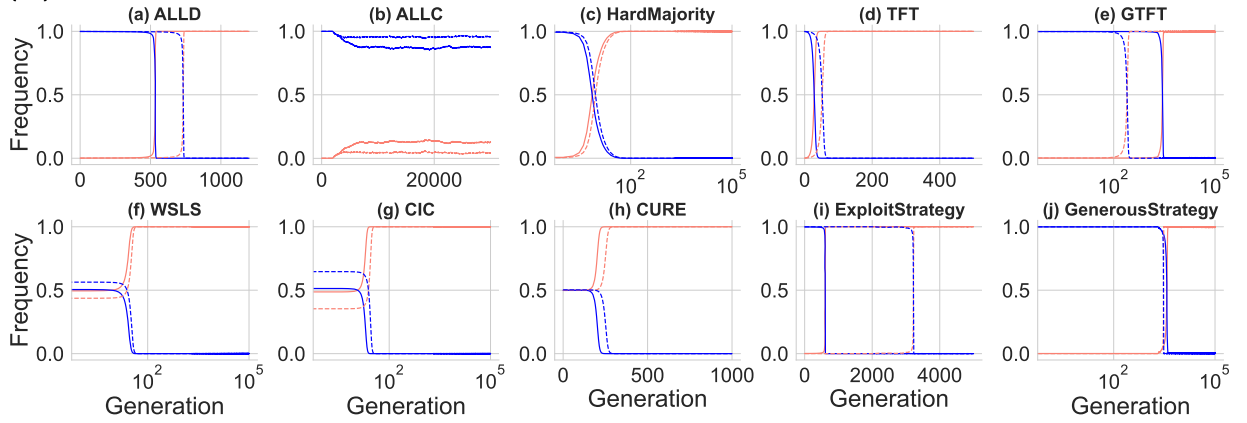
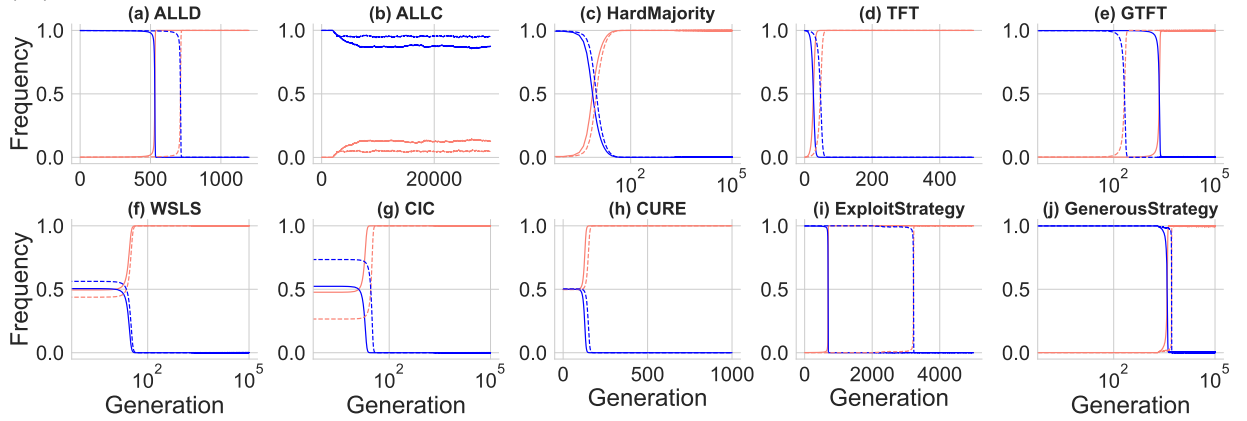
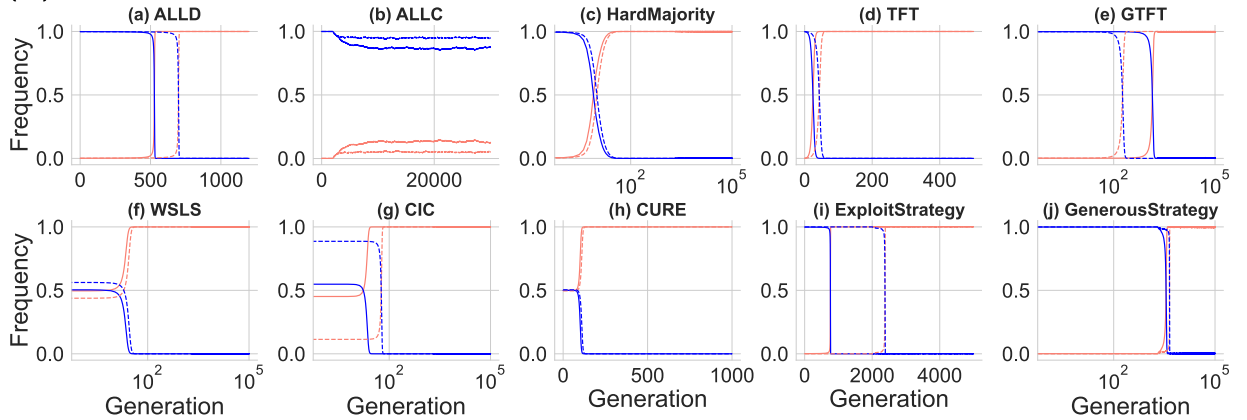
(A) $m = 3$ **(B) $m = 5$** **(C) $m = 10$** 

Fig. A. 2. Two-strategy populations of iterated stag hunt dilemma, where one strategy is MCSUC, with $m = 3$ channels (A), $m = 5$ channels (B), and $m = 10$ channels (C). There are two error rates: $\epsilon = 1\%$ (solid line) and $\epsilon = 10\%$ (dashed line). The red line represents MCSUC, while the blue line represents the opponent's strategy. The initial frequency of MCSUC is 0.1% in most populations and remains below 50% in others. MCSUC successfully invades nine out of the ten strategies except for ALLC. This indicates that MCSUC strategy is evolutionarily more advantaged than nine of the ten strategies, except for the ALLC strategy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

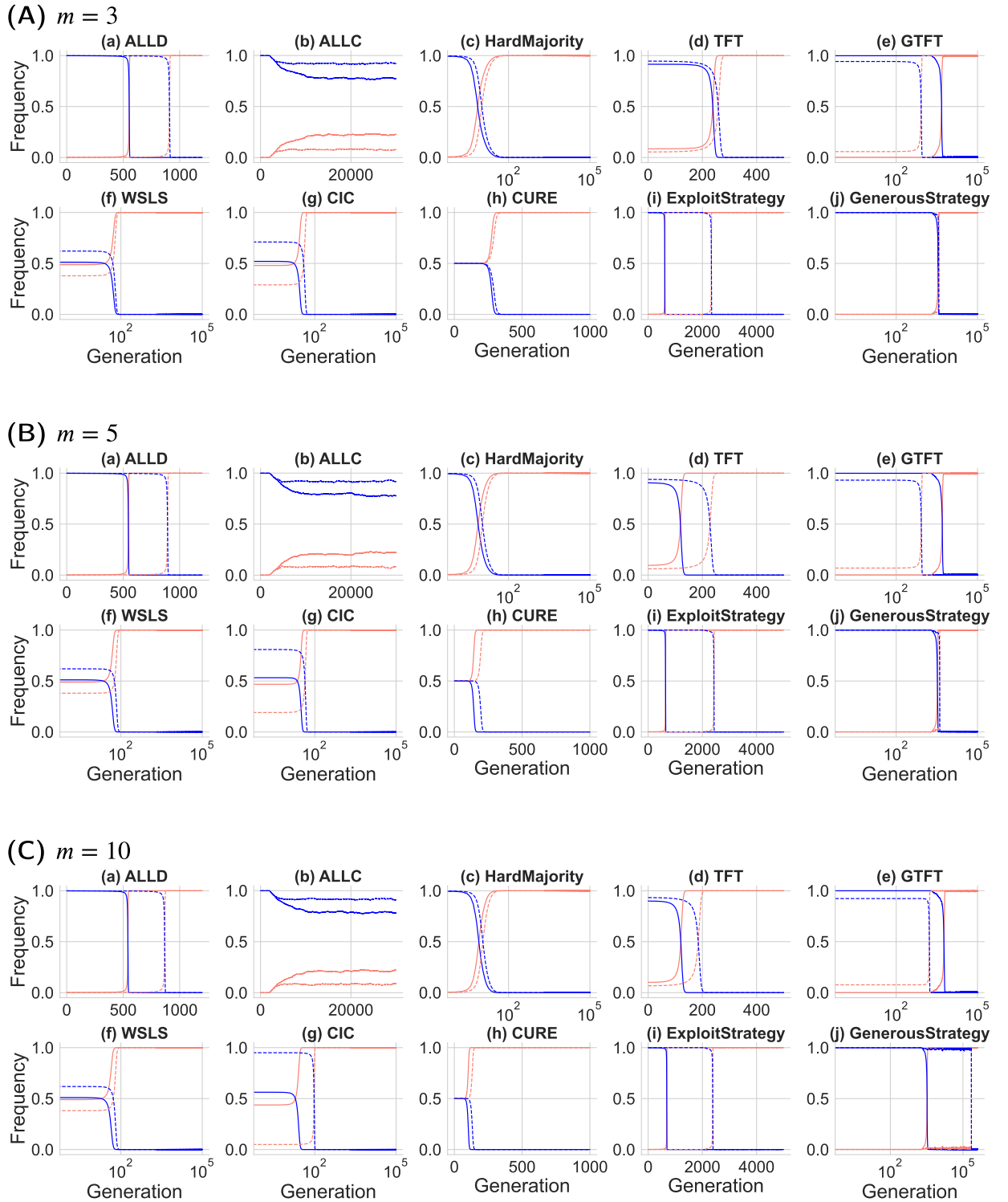


Fig. A. 3. Two-strategy populations of iterated snowdrift dilemma, where one strategy is MCSUC, with $m = 3$ channels (A), $m = 5$ channels (B), and $m = 10$ channels (C). There are two error rates: $\epsilon = 1\%$ (solid line) and $\epsilon = 10\%$ (dashed line). The red line represents MCSUC, while the blue line represents the opponent's strategy. The initial frequency of MCSUC is 0.1% in most populations and remains below 50% in others. MCSUC successfully invades nine out of the ten strategies except for ALLC. This indicates that MCSUC strategy is evolutionarily more advantaged than nine of the ten strategies, except for the ALLC strategy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix B. Weak performance of WSLs & CIC

The multichannel-WSLS strategy dictates that an agent cooperates in channel k only if both the agent and its opponent chose the same action in that channel during the previous round; otherwise, the agent defects in the next round for that channel. Similarly, the CIC strategy specifies that an agent cooperates across all channels only if both the agent and its opponent chose the same actions across all channels in the previous round. Any mismatch in actions results in defection across all channels in the following round.

Consider an agent X using either the multichannel-WSLS or CIC strategy in an m -channel IPD. Suppose that in a given round t , both X and its opponent defect. In the next round $t + 1$, X will return to cooperation regardless of whether the opponent continues to defect. This means that even if the opponent consistently defects, X will repeatedly revert to cooperation after one round of defection. As a result, X is vulnerable to exploitation by an opponent who always defects, as X fails to sustain retaliation and thus will be continuously exploited.

Inspired by this observation, we hypothesize that these two strategies are vulnerable to long-lasting exploitation, and may easily lose to exploitative strategies. To test this hypothesis, we conduct experiments and pair a multichannel-WSLS or CIC agent with an opponent that adopts one of the three types of exploitative strategies: the ALLD strategy, HardMajority, and the extortion strategy. Note that the extortion strategy involves a large class of strategies; we select two of them that stand for different levels of extortion. Each experiment was run for 1,000,000 rounds with the number of channels set to 3, 5, and 10. The average payoff for each strategy under $\epsilon = 1\%$ is presented in the main text (Fig. 1), where we demonstrated the vulnerability of multichannel-WSLS and CIC to long-term exploitation by their exploitative opponents.

In this section, we include additional results for experiments conducted without error ($\epsilon = 0\%$) in Fig. B. 1. The results are broadly consistent with those presented in the main text, further confirming the susceptibility of multichannel-WSLS and CIC strategies to long-lasting exploitation.

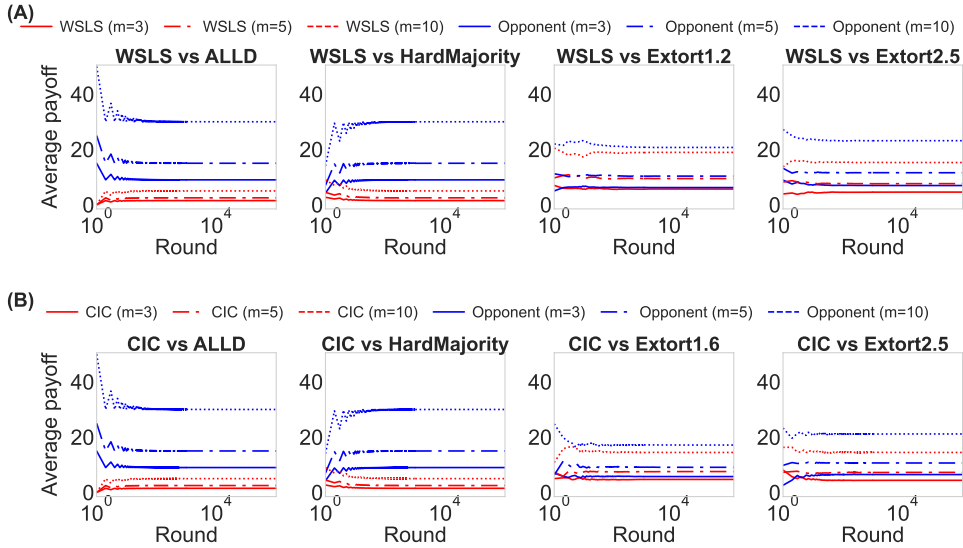


Fig. B. 1. Two-agent experiments were conducted under three different channel settings ($m = 3, 5, 10$). In these experiments, one agent adopts the multichannel-WSLS or CIC strategy, while the opponent employs the ALLD, HardMajority, or extortion strategies. The error rate is fixed at $\epsilon = 0\%$. The top row shows WSLs's expected payoffs against the four opponents (ALLD, HardMajority, Extort1.2, Extort2.5), and the bottom row shows CIC's expected payoffs against the same opponents. In each subplot, red curves represent WSLs (or CIC), and blue curves represent the opponent; the line styles (solid, dash-dotted, dashed) correspond to $m = 3, 5, 10$ channels, respectively. In each experiment, both the multichannel-WSLS and CIC tend to be exploited in the long term by the ALLD, HardMajority, and the extortion strategies. The error rate $\epsilon = 0\%$. The payoff matrix is set as $R_k = 3$, $S_k = 0$, $T_k = 5$ and $P_k = 1$ for each channel k . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix C. Detailed proofs of analytical results

C.1. Proof for Theorem 1

Theorem 1. In an m -channel IPD, if agents X and Y both adopt the MCSUC strategy, will always cooperate in all channels, leading to full cooperation. Their expected cooperation rates are $\rho_X = \rho_Y = 1$, and the expected payoff are $r_X = r_Y = R_1 + R_2 + \dots + R_m$.

Proof. In an m -channel IPD, under the self-play setting, let $\Delta_{X,1}, \Delta_{X,2}, \dots, \Delta_{X,m}$ be agent X 's tolerance thresholds, and let $\Delta_{Y,1}, \Delta_{Y,2}, \dots, \Delta_{Y,m}$ be agent Y 's tolerance thresholds. Initially, at $t = 1$ round, $n_X(1) = n_Y(1) = 0$, since no interaction (and thus no defection) occurs prior to that. For agent X , according to Definition 1, since $d_X(1) = n_Y(1) - n_X(1) = 0 \leq \Delta_{X,1}$, agent X will cooperate on both

channels at $t = 1$ round. Similarly, for agent Y , the condition $d_Y(1) = n_X(1) - n_Y(1) = 0 \leq \Delta_{Y,1}$ is satisfied, Y will choose cooperation on both channels at $t = 1$ round too. The full cooperation outcome at $t = 1$ round ensures $n_X(2) = n_Y(2) = 0$ i.e., $d_X(2) = 0 \leq \Delta_{X,1}$ and $d_Y(2) = 0 \leq \Delta_{Y,1}$ at $t = 2$ round, which results in the same full cooperation outcome at $t = 2$ round. Repeating this manner infinitely, agents X and Y achieve full cooperation throughout the entire game. Therefore, their expected cooperation rates are $\rho_X = \rho_Y = 1$, and the expected payoffs are $r_X = r_Y = R_1 + R_2 + \dots + R_m$. $\square$

C.2. Proof for Proposition 1

Proposition 1. In an m -channel IPD with an error rate ϵ ($0 \leq \epsilon < 0.5$), the expected cooperation rates for agents X and Y that both use MCSUC are

$$\rho_X = \rho_Y \approx 1 - \epsilon + \frac{\epsilon - 2\epsilon^2}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1}, \quad (\text{C. 1})$$

which monotonically increases as the error rate ϵ decreases, or as the tolerance thresholds $\Delta_{X,1}, \Delta_{Y,1}$ increase.

Proof. For an arbitrary round t , we consider $x_{d_X}(t)$ as the probability at the t round, the difference in the cumulative number of defections between two agents accumulated over the previous $t - 1$ rounds is $d_X = n_Y(t) - n_X(t)$. Assume $x_{d_X}(t)$ is known for all $d_X \in \mathbb{Z}$, then we can determine the difference probability of next round $x_{d_X}(t + 1)$:

$$\begin{aligned} & \vdots \\ x_{-\Delta_{Y,1}}(t + 1) &= \left(\sum_{k=0}^m \binom{m}{k}^2 \epsilon^{2k} (1 - \epsilon)^{2(m-k)} \right) x_{-\Delta_{Y,1}}(t) + \left(\sum_{k=0}^{m-1} \binom{m}{k+1} \binom{m}{k} \epsilon^{2k+1} (1 - \epsilon)^{2m-2k-1} \right) x_{-\Delta_{Y,1}+1}(t) + \left((1 - \epsilon)^{2m} + \dots \right) x_{-\Delta_{Y,1}-1}(t) + \dots \\ x_{-\Delta_{Y,1}+1}(t + 1) &= \left(\sum_{k=0}^m \binom{m}{k}^2 \epsilon^{2k} (1 - \epsilon)^{2(m-k)} \right) x_{-\Delta_{Y,1}+1}(t) + \left(\sum_{k=0}^{m-1} \binom{m}{k+1} \binom{m}{k} \epsilon^{2k+1} (1 - \epsilon)^{2m-2k-1} \right) x_{-\Delta_{Y,1}+2}(t) \\ &+ \left(\sum_{k=1}^m \binom{m}{k-1} \binom{m}{k} \epsilon^{2k-1} (1 - \epsilon)^{2m-2k+1} \right) x_{-\Delta_{Y,1}}(t) + \dots \\ & \vdots \\ x_i(t + 1) &= \left(\sum_{k=0}^m \binom{m}{k}^2 \epsilon^{2k} (1 - \epsilon)^{2(m-k)} \right) x_i(t) + \left(\sum_{k=0}^{m-1} \binom{m}{k+1} \binom{m}{k} \epsilon^{2k+1} (1 - \epsilon)^{2m-2k-1} \right) x_{i+1}(t) + \left(\sum_{k=1}^m \binom{m}{k-1} \binom{m}{k} \epsilon^{2k-1} (1 - \epsilon)^{2m-2k+1} \right) x_{i-1}(t) + \dots \text{for } -\Delta_{Y,1} + \\ & m \leq i \leq \Delta_{X,1} - m \\ & \vdots \\ x_{\Delta_{X,1}-1}(t + 1) &= \left(\sum_{k=0}^m \binom{m}{k}^2 \epsilon^{2k} (1 - \epsilon)^{2(m-k)} \right) x_{\Delta_{X,1}-1}(t) + \left(\sum_{k=0}^{m-1} \binom{m}{k+1} \binom{m}{k} \epsilon^{2k+1} (1 - \epsilon)^{2m-2k-1} \right) x_{\Delta_{X,1}}(t) + \left(\sum_{k=1}^m \binom{m}{k-1} \binom{m}{k} \epsilon^{2k-1} (1 - \epsilon)^{2m-2k+1} \right) x_{\Delta_{X,1}-2}(t) + \dots \\ x_{\Delta_{X,1}}(t + 1) &= \left(\sum_{k=0}^m \binom{m}{k}^2 \epsilon^{2k} (1 - \epsilon)^{2(m-k)} \right) x_{\Delta_{X,1}}(t) + \left((1 - \epsilon)^{2m} + \dots \right) x_{\Delta_{X,1}+1}(t) + \left(\sum_{k=1}^m \binom{m}{k-1} \binom{m}{k} \epsilon^{2k-1} (1 - \epsilon)^{2m-2k+1} \right) x_{\Delta_{X,1}-1}(t) + \dots \\ & \vdots \end{aligned} \quad (\text{C. 2})$$

For all rounds $t > 0$, these equations hold. By summing up the corresponding equations for t within the range of $\{1, \dots, T\}$, dividing both sides by T , and then approaching the limit as $T \rightarrow \infty$, approximately treating the squared term of ϵ in the transition probability as negligible, we arrive at:

$$\begin{aligned} x_{-\Delta_{Y,1}} &= (1 - 2m\epsilon)x_{-\Delta_{Y,1}} + (m\epsilon)x_{-\Delta_{Y,1}+1} + (1 - 2m\epsilon)x_{-\Delta_{Y,1}-1} \\ x_{-\Delta_{Y,1}+1} &= (1 - 2m\epsilon)x_{-\Delta_{Y,1}+1} + (m\epsilon)x_{-\Delta_{Y,1}+2} + (m\epsilon)x_{-\Delta_{Y,1}} + \epsilon x_{-\Delta_{Y,1}-1} \\ & \vdots \\ x_i &= (1 - 2m\epsilon)x_i + (m\epsilon)x_{i+1} + (m\epsilon)x_{i-1} \quad \text{for } -\Delta_{Y,1} + 2 \leq i \leq \Delta_{X,1} - 2 \\ & \vdots \\ x_{\Delta_{X,1}-1} &= (1 - 2m\epsilon)x_{\Delta_{X,1}-1} + (m\epsilon)x_{\Delta_{X,1}} + \epsilon x_{\Delta_{X,1}+1} + (m\epsilon)x_{\Delta_{X,1}-2} \\ x_{\Delta_{X,1}} &= (1 - 2m\epsilon)x_{\Delta_{X,1}} + (1 - 2m\epsilon)x_{\Delta_{X,1}+1} + (m\epsilon)x_{\Delta_{X,1}-1} \end{aligned} \quad (\text{C. 3})$$

where $x_{d_X} := \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T x_{d_X}(t)$ is the average time to observe d_X over all rounds t . Because this is a self-play of our strategy, it is symmetrical with respect to $d_X = (\Delta_{X,1} - \Delta_{Y,1})/2$ for all $d_X \in \mathbb{N}$. Therefore, we get:

$$\begin{aligned} x_i &= (1 - 2m\epsilon)x_i + (2m\epsilon)x_{i+1} \quad \text{for } (\Delta_{X,1} - \Delta_{Y,1})/2 \leq i \leq \Delta_{X,1} - 2 \\ & \vdots \\ x_{\Delta_{X,1}-1} &= (1 - 2m\epsilon)x_{\Delta_{X,1}-1} + (m\epsilon)x_{\Delta_{X,1}} + (m-1)\epsilon x_{\Delta_{X,1}+1} + (m\epsilon)x_{\Delta_{X,1}-2} \\ x_{\Delta_{X,1}} &= (1 - 2m\epsilon)x_{\Delta_{X,1}} + (1 - 2m\epsilon)x_{\Delta_{X,1}+1} + (m\epsilon)x_{\Delta_{X,1}-1} \end{aligned} \quad (\text{C. 4})$$

Solving each equation, we obtain:

$$\begin{aligned} x_i &= x_{i+1} \quad \text{for } (\Delta_{X,1} - \Delta_{Y,1})/2 \leq i \leq \Delta_{X,1} - 2 \\ &\vdots \\ x_{\Delta_{X,1}} &= \frac{m\epsilon + \epsilon - 1}{2\epsilon - 1} x_{\Delta_{X,1}-1} \\ x_{\Delta_{X,1}+1} &= \frac{m\epsilon}{1 - m\epsilon - \epsilon} x_{\Delta_{X,1}} \end{aligned} \quad (\text{C. 5})$$

Utilizing the condition that the sum of all x_{d_X} equals 1,

$$1 = \sum_{d_X \in \mathbb{Z}} x_{d_X}, \quad (\text{C. 6})$$

for $(\Delta_{X,1} - \Delta_{Y,1})/2 \leq i \leq \Delta_{X,1} - 1$, we get

$$x_i = \frac{2\epsilon - 1}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1}. \quad (\text{C. 7})$$

To calculate the average cooperation rates of the agents, we observe the following: When $-\Delta_{Y,1} \leq d_X \leq \Delta_{X,1}$, the expected cooperation rate of two agents across m channels is $1 - \epsilon$. For the case when $\Delta_{X,1} < d_X \leq \Delta_{X,2}$ or $-\Delta_{Y,2} \leq d_X < -\Delta_{Y,1}$, the expected cooperation rate of two agents across m channels is $1 - \epsilon + \frac{2\epsilon-1}{2m}$. Then based on x_{d_X} , the cooperation rates of agent X and Y can be calculated as:

$$\rho_X = \rho_Y \approx 1 - \epsilon + \frac{\epsilon - 2\epsilon^2}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1}, \quad (\text{C. 8})$$

We can find the derivative of the cooperation rate ρ_X with respect to ϵ to be:

$$\frac{d\rho_X}{d\epsilon} = - \frac{(-1 + 2\epsilon)(1 + \Delta_{X,1} + \Delta_{Y,1})(-2 + (-1 + 2\epsilon)(\Delta_{X,1} + \Delta_{Y,1}))}{(-1 + (-1 + 2\epsilon)(\Delta_{X,1} + \Delta_{Y,1}))^2} \quad (\text{C. 9})$$

Then, we can determine that $\frac{d\rho_X}{d\epsilon} < 0$ for $\epsilon \in (0, 0.5)$.

Then we calculated the expected payoff of agents X and Y in a similar process. The expected payoff of agent X and Y is:

$$\begin{aligned} r_X = r_Y \approx & \left\{ \left[\frac{2m\epsilon}{1 - 2\epsilon} + \Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right] \sum_{k=1, k \neq p_1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] + \right. \\ & \left(\Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right) [(1 - \epsilon)^2 R_{p_1} + \epsilon(1 - \epsilon)(T_{p_1} + S_{p_1}) + \epsilon^2 P_{p_1}] + \\ & \left. \frac{m\epsilon}{1 - 2\epsilon} [2\epsilon(1 - \epsilon)(R_{p_1} + P_{p_1}) + (1 - 2\epsilon + 2\epsilon^2)(S_{p_1} + T_{p_1})] \right\} \left(\frac{2\epsilon - 1}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1} \right) \end{aligned} \quad (\text{C. 10})$$

where p_1 here represents the channel number with the highest defection priority. To allow for a better understanding of the effects of the error rate and tolerance thresholds, we illustrate the expected payoff in a specific case of the prisoner's dilemma—the donation game (Sigmund 2010). Then we get

$$\begin{aligned} r_X = r_Y \approx & \left\{ \left[\frac{2m\epsilon}{1 - 2\epsilon} + \Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right] (m - 1)[(1 - \epsilon)(b - c)] + \right. \\ & \left(\Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right) [(1 - \epsilon)(b - c)] + \\ & \left. \frac{m\epsilon}{1 - 2\epsilon} [2\epsilon(1 - \epsilon)(b - c)] \right\} \left(\frac{2\epsilon - 1}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1} \right) \end{aligned} \quad (\text{C. 11})$$

which is the same as the cooperation rate equation except $(b - c)$, they have the same trend when ϵ , $\Delta_{X,1}$ or $\Delta_{Y,1}$ change.

We can find the derivative of the payoff r_X with respect to ϵ to be:

$$\begin{aligned} \frac{dr_X}{d\epsilon} = & -((b - c)m[(1 - 2\epsilon)(3 - 2\epsilon) + 2\epsilon^2 + 2((1 - 2\epsilon)(2 - 2\epsilon) + (3 - 4\epsilon)\epsilon^2)(\Delta_{X,1} + \Delta_{Y,1}) \\ & + (1 - 2\epsilon)^2(\Delta_{X,1} + \Delta_{Y,1})^2] / [(-1 + (-1 + 2\epsilon)(\Delta_{X,1} + \Delta_{Y,1}))^2]) \end{aligned} \quad (\text{C. 12})$$

According to the definition of the donation game,

$$b - c > 0, \quad (\text{C. 13})$$

and there is an error, i.e., $0 < \epsilon < 0.5$ and $(\Delta_{X,1} + \Delta_{Y,1}) \geq 2$, then

$$b - c > 0, \quad (\text{C. 14})$$

this implies

$$(1 - 2\epsilon)(3 - 2\epsilon) + 2\epsilon^2 + 2((1 - 2\epsilon)(2 - 2\epsilon) + (3 - 4\epsilon)\epsilon^2)(\Delta_{X,1} + \Delta_{Y,1}) + (1 - 2\epsilon)^2(\Delta_{X,1} + \Delta_{Y,1})^2 > 0, \quad (\text{C. 15})$$

$$(-1 + (-1 + 2\epsilon)(\Delta_{X,1} + \Delta_{Y,1}))^2 > 0, \quad (\text{C. 16})$$

and

$$m \geq 1. \quad (\text{C. 17})$$

Then, we can determine that $\frac{dr_X}{d\epsilon} < 0$ for $\epsilon \in (0, 0.5)$, and based on the equal payoff of agent X and Y , we can conclude that the cooperation rates and payoff exhibits a monotonically declining trend in relation to the error rate ϵ .

At the same time, we can find the derivative of the payoff r_X with respect to $\Delta_{X,1}$ or $\Delta_{Y,1}$ to be:

$$\frac{dr_X}{d\Delta_{X,1}} = \frac{dr_X}{d\Delta_{Y,1}} = \frac{2(b-c)m(-1+\epsilon)^2\epsilon(1-2\epsilon)}{(-1+(\Delta_{X,1}+\Delta_{Y,1})(-1+2\epsilon))^2}, \quad (\text{C. 18})$$

there is an error, i.e., $0 < \epsilon < 0.5$, $m \geq 1$ and $(\Delta_{X,1} + \Delta_{Y,1}) \geq 2$, then

$$1 - 2\epsilon > 0, \quad (\text{C. 19})$$

this implies

$$2(b-c)m(-1+\epsilon)^2\epsilon(1-2\epsilon) > 0, \quad (\text{C. 20})$$

$$(-1 + (\Delta_{X,1} + \Delta_{Y,1})(-1 + 2\epsilon))^2 > 0. \quad (\text{C. 21})$$

Then, we can determine $\frac{dr_X}{d\Delta_{X,1}} = \frac{dr_X}{d\Delta_{Y,1}} > 0$ for $\epsilon \in (0, 0.5)$, and based on the equal payoff of agent X and Y , we can conclude that the cooperation rates and payoff monotonically increase as the tolerance thresholds increase. $\square$

C.3. Proof for Theorem 2

Theorem 2. In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses the ALLD strategy. With a positive error rate ϵ ($0 < \epsilon < 0.5$), both agents have the same expected cooperation rate $\rho_X = \rho_Y = \epsilon$, and have the same expected payoff

$$r_X = r_Y = \epsilon^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + (1 - \epsilon)^2(P_1 + P_2 + \dots + P_m). \quad (\text{C. 22})$$

Proof. We'll use a proof by contradiction to demonstrate our theorem. It is known that the average number of total defection times across m channels for agent Y is: $\lim_{T \rightarrow \infty} \frac{n_Y(T)}{T} = m(1 - \epsilon)$. Assume the difference between the average number of total defection times of agent Y and X $\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T} - \frac{n_X(T)}{T}) > 0$. Then there exists some t_0 such that $d_X(t) > \Delta_{X,m}$ for all $t > t_0$, our strategy will defect on all channels for all but finitely many rounds. Then our strategy's average number of total defection times equals $m(1 - \epsilon)$, the maximum average number of total defection times, which contradicts the assumption. So $\lim_{T \rightarrow \infty} (\frac{n_X(T)}{T}) \geq m(1 - \epsilon)$ is satisfied, at the same time, the average total number of defection times of our strategy cannot exceed the theoretical maximum of $m(1 - \epsilon)$. Therefore, our strategy's average number of total defection times must be $m(1 - \epsilon)$, indicating that when our strategy competes against ALLD in a multichannel iterated PD, it behaves like an ALLD strategy.

When error rate $\epsilon > 0$, the average cooperation obtained from self-play of ALLD is ϵ , and the average payoffs is:

$$r_X = r_Y = \epsilon^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + (1 - \epsilon)^2(P_1 + P_2 + \dots + P_m). \quad (\text{C. 23})$$

$\square$

C.4. Proof for Theorem 3

Theorem 3. In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses the ALLC strategy. With a positive error rate ϵ ($0 < \epsilon < 0.5$), both agents have the same expected cooperation rate $\rho_X = \rho_Y = 1 - \epsilon$, and have the same expected payoff

$$r_X = r_Y = (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m). \quad (\text{C. 24})$$

Proof. Consider a similar proof method as Theorem 2, assume that the average number of total defection times of ALLC is lower than that of our strategy, i.e., $\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T} - \frac{n_X(T)}{T}) < 0$. Then there must exist some t_0 such that $d_X(t) < \Delta_{X,1}$ for all $t > t_0$, our strategy will cooperate on all channels for all but finitely many rounds. Then our strategy's average number of total defection times equals $m\epsilon$, the minimum average number of total defection times, which contradicts the assumption. So $\lim_{T \rightarrow \infty} (\frac{n_X(T)}{T}) \leq m\epsilon$ is satisfied, at the same time, the average total number of defection times of our strategy cannot be less than the theoretical minimum of $m\epsilon$. Therefore,

our strategy's average total number of defection times must be $m\epsilon$, indicating that when our strategy competes against ALLC in a multichannel iterated PD, it behaves like an ALLC strategy. So the expected payoff is calculated by the self-play of the ALLC strategy.

When error rate $\epsilon > 0$, the average cooperation obtained from self-play of ALLC is $1 - \epsilon$, and the average payoffs is:

$$r_X = r_Y = (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m). \quad (C. 25)$$

□

C.5. Proof for Theorem 4

Theorem 4. In an m -channel IPD, suppose agent X uses MCSUC while opponent agent Y uses a strategy α . For any opponent strategy α , both agents always have the same expected cooperation rate $\rho_X = \rho_Y$. Moreover, if the m -channel IPD satisfies $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$, both agents also have the same expected payoff $r_X = r_Y$.

Proof. We'll use a proof by contradiction to demonstrate our theorem. Assume the difference between the average number of total defection times of agent Y and X $\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T} - \frac{n_X(T)}{T}) > 0$. Then there exist some t_0 such that $d_X(t) > \Delta_{X,m}$ for all $t > t_0$, our strategy will defect on all channels for all but finitely many rounds. Then our strategy's average number of total defection times equals $m(1 - \epsilon)$, the maximum average number of total defection times. Because $\lim_{T \rightarrow \infty} \frac{n_Y(T)}{T} \leq m(1 - \epsilon)$, which contradicts the assumption. The assumption $\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T} - \frac{n_X(T)}{T}) < 0$ yields a similar contradiction. So $\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T} - \frac{n_X(T)}{T}) = 0$ is satisfied. We get:

$$\lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T}) = \lim_{T \rightarrow \infty} (\frac{n_X(T)}{T}), \quad (C. 26)$$

is satisfied too. Since

$$\rho_X = 1 - \frac{1}{m} \lim_{T \rightarrow \infty} (\frac{n_X(T)}{T}) = \rho_Y = 1 - \frac{1}{m} \lim_{T \rightarrow \infty} (\frac{n_Y(T)}{T}) \quad (C. 27)$$

, so $\rho_X = \rho_Y$.

Additionally, we can calculate the expected payoff of agent X , first we set $a_X^k(t)$ as indicator function, at round t , if agent X choose cooperation in k channel, $a_X^k(t) = 1$, if agent Y choose cooperation in k channel, $a_Y^k(t) = 1$, then the expected payoff of X is:

$$\begin{aligned} r_X = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t & (a_X^1(t)a_Y^1(t)R_1 + a_X^1(t)(1 - a_Y^1(t))S_1 + (1 - a_X^1(t))a_Y^1(t)T_1 + \\ & (1 - a_X^1(t))(1 - a_Y^1(t))P_1 + a_X^2(t)a_Y^2(t)R_2 + a_X^2(t)(1 - a_Y^2(t))S_2 + (1 - a_X^2(t))a_Y^2(t)T_2 + \\ & (1 - a_X^2(t))(1 - a_Y^2(t))P_2 + \dots + a_X^m(t)a_Y^m(t)R_m + a_X^m(t)(1 - a_Y^m(t))S_m + \\ & (1 - a_X^m(t))a_Y^m(t)T_m + (1 - a_X^m(t))(1 - a_Y^m(t))P_m). \end{aligned} \quad (C. 28)$$

Similarly, the expected payoff of Y is:

$$\begin{aligned} r_Y = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t & (a_X^1(t)a_Y^1(t)R_1 + a_X^1(t)(1 - a_Y^1(t))T_1 + (1 - a_X^1(t))a_Y^1(t)S_1 + \\ & (1 - a_X^1(t))(1 - a_Y^1(t))P_1 + a_X^2(t)a_Y^2(t)R_2 + a_X^2(t)(1 - a_Y^2(t))T_2 + (1 - a_X^2(t))a_Y^2(t)S_2 + \\ & (1 - a_X^2(t))(1 - a_Y^2(t))P_2 + \dots + a_X^m(t)a_Y^m(t)R_m + a_X^m(t)(1 - a_Y^m(t))T_m + \\ & (1 - a_X^m(t))a_Y^m(t)S_m + (1 - a_X^m(t))(1 - a_Y^m(t))P_m). \end{aligned} \quad (C. 29)$$

The difference of r_Y and r_X is:

$$\begin{aligned} r_Y - r_X = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t & (a_X^1(t)(1 - a_Y^1(t))T_1 + (1 - a_X^1(t))a_Y^1(t)S_1 + a_X^2(t)(1 - a_Y^2(t))T_2 + \\ & (1 - a_X^2(t))a_Y^2(t)S_2 + \dots + a_X^m(t)(1 - a_Y^m(t))T_m + (1 - a_X^m(t))a_Y^m(t)S_m) - \\ & \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t (a_X^1(t)(1 - a_Y^1(t))S_1 + (1 - a_X^1(t))a_Y^1(t)T_1 + \\ & + a_X^2(t)(1 - a_Y^2(t))S_2 + (1 - a_X^2(t))a_Y^2(t)T_2 + \dots + a_X^m(t)(1 - a_Y^m(t))S_m + (1 - a_X^m(t))a_Y^m(t)T_m) \\ = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t & ((a_X^1(t) - a_Y^1(t))T_1 - (a_X^1(t) - a_Y^1(t))S_1 + (a_X^2(t) - a_Y^2(t))T_2 - \\ & (a_X^2(t) - a_Y^2(t))S_2 + \dots + (a_X^m(t) - a_Y^m(t))T_m - (a_X^m(t) - a_Y^m(t))S_m) \\ = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_t & ((T_1 - S_1)(a_X^1(t) - a_Y^1(t)) + (T_2 - S_2)(a_X^2(t) - a_Y^2(t)) + \dots + (T_m - S_m)(a_X^m(t) - a_Y^m(t))) \\ = (T_1 - S_1) \lim_{T \rightarrow \infty} & (\frac{n_Y^1(T)}{T} - \frac{n_X^1(T)}{T}) + (T_2 - S_2) \lim_{T \rightarrow \infty} (\frac{n_Y^2(T)}{T} - \frac{n_X^2(T)}{T}) + \dots + (T_m - S_m) \lim_{T \rightarrow \infty} (\frac{n_Y^m(T)}{T} - \frac{n_X^m(T)}{T}) \end{aligned} \quad (C. 30)$$

$$= \sum_{k=1}^m ((T_k - S_k) (\lim_{T \rightarrow \infty} (\frac{n_Y^i(T)}{T} - \frac{n_X^i(T)}{T}))).$$

If the m -channel IPD satisfies $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$, combined with the already proven formula (C. 26), we can conclude that both agents also have the same expected payoff $r_X = r_Y$. $\square$

C.6. Proof for Theorem 5

Theorem 5. In an m -channel IPD, suppose agent X uses MCSUC while the opponent agent Y uses a strategy α . For any opponent strategy α , if condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied, $r_X \geq r_Y$ is ensured.

Proof. Based on Theorem 4, it is known that

$$r_Y - r_X = \sum_{k=1}^m ((T_k - S_k) (\lim_{T \rightarrow \infty} (\frac{n_Y^i(T)}{T} - \frac{n_X^i(T)}{T}))) \quad (\text{C. 31})$$

If $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is satisfied, combined with the already proven formula (C. 26), agent X will obtain the same expected payoff as agent Y . Otherwise, if $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied, set the defection priority to be consistent with the size relationship between $(T_k - S_k)$ across the m channels. Based on the already proven formula (C. 26), agent x will prioritize defecting in the channels with higher values of $(T_k - S_k)$, choosing to defect as much as possible according to the order of $(T_k - S_k)$ in different channels, then $r_X \geq r_Y$ can be ensured. $\square$

C.7. Proof for Corollary 1

Corollary 1. In an m -channel IPD, suppose agent X adopts MCSUC while the opponent agent Y uses a strategy α . For any opponent strategy α , there always exists a priority vector $\mathbf{p}^T$ such that the expected payoff of our strategy is not smaller than that of the opponent strategy, i.e. $r_X \geq r_Y$.

Proof. Putting Theorems 4 and 5 together, when condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is satisfied, agent X will obtain the same expected payoff as agent Y , i.e. $r_X = r_Y$. When condition $(T_1 - S_1) = (T_2 - S_2) = \dots = (T_m - S_m)$ is not satisfied, by setting the defection priority p to be consistent with the size relationship between $(T_k - S_k)$ across the m channels, we can ensure our strategy never loses to any opponent strategy in terms of the expected payoff, i.e. $r_X \geq r_Y$. So we can conclude that for any opponent strategy α , there always exists a p (determines the defection priority) such that the expected payoff of our strategy is not smaller than that of the opponent strategy, i.e. $r_X \geq r_Y$. $\square$

C.8. Proof for Theorem 6

Theorem 6. In an m -channel IPD without the presence of errors, the strategy profile $(\text{MCSUC}, \text{MCSUC})$ is a Nash equilibrium as well as a subgame perfect equilibrium.

Proof. To show the result, we need to show that no deviation from $\text{MCSUC}_{\Delta_2, \Delta_1, \dots, \Delta_m}$ is profitable, after no previous history of the game. To this end, consider an arbitrary history such that the current round's defection difference statistic is d .

Then the continuation payoff of a focal agent who continues to use $\text{MCSUC}_{\Delta_1, \Delta_2, \dots, \Delta_m}$ is $R_1 + R_2 + \dots + R_m$. This follows from the observation that independent of the current value of d , two $\text{MCSUC}_{\Delta_1, \Delta_2, \dots, \Delta_m}$ agents will reach $|d| \leq \Delta_1$ in finite time with certainty. From then on, the agents' continuation payoff will be given by $R_1 + R_2 + \dots + R_m$; the finitely many rounds before that do not affect the agents' long-run payoffs.

Now suppose there is a deviation such that the focal agent obtains a continuation payoff larger than $R_1 + R_2 + \dots + R_m$. Then it follows from Corollary 1 that also the other agent gets a payoff larger than $R_1 + R_2 + \dots + R_m$ (since the other agent continues to adopt $\text{MCSUC}_{\Delta_1, \Delta_2, \dots, \Delta_m}$). This implies that the joint payoff of both agents is larger than $2(R_1 + R_2 + \dots + R_m)$, which contradicts the assumption that $2(R_1 + R_2 + \dots + R_m)$ is the maximum payoff that is achievable in any single round. $\square$

C.9. Proof for Proposition 2

Proposition 2. In an m -channel IPD with a positive error rate ϵ ($0 < \epsilon < 0.5$), the strategy profile $(\text{MCSUC}, \text{MCSUC})$ is not a Nash equilibrium.

Proof. Based on Proposition 1 and Theorem 3, in an m -channel IPD with a positive error rate $\epsilon > 0$, under the self-play setting of our proposed strategy with an error rate ϵ , each agent's expected cooperation rate is

$$\rho_X = \rho_Y \approx 1 - \epsilon + \frac{\epsilon - 2\epsilon^2}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1}, \quad (\text{C. 32})$$

if one agent deviates from our proposed strategy to ALLC strategy, the expected cooperation rate of the deviated agent and the other agent with our proposed strategy will be $\rho_X = \rho_Y = 1 - \epsilon$, which equals the highest expected cooperation rate of strategy profile $(\text{ALLC}, \text{ALLC})$.

Then it has been proven in [Proposition 1](#) that $\frac{\epsilon - 2\epsilon^2}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1} < 0$, under the self-play setting of our proposed strategy with an error rate ϵ , the expected payoff for each agent is lower than the expected payoff from self-play with an ALLC strategy.

Therefore, under the self-play setting of our proposed strategy with an error rate ϵ , if an agent deviates from our proposed strategy to the ALLC strategy, both the deviated agent and the agent that continues to maintain our proposed strategy will receive the expected cooperation rate and expected payoff of ALLC in self-play. This means that the agent gains a higher payoff by deviating from our proposed strategy to ALLC, which implies that the strategy profile $(MCSUC, MCSUC)$ is not a Nash equilibrium. $\square$

C.10. Proof for [Theorem 7](#)

Theorem 7. In an m -channel IPD with a positive error rate ϵ ($0 < \epsilon < 0.5$), given a non-negative $\hat{\epsilon}$, the strategy profile $(MCSUC, MCSUC)$ is an $\hat{\epsilon}$ -equilibrium if $\hat{\epsilon}$ and for all $\epsilon \leq \frac{3\hat{\epsilon}}{m \max \left[(R_{p_1} - P_{p_1}), (2R_{p_1} - S_{p_1} - T_{p_1}) \right]}$

Proof. The maximum payoff that two agents can jointly achieve in an IPD with errors is twice the payoff that they get if they both intend to cooperate in every round,

$$\pi(\text{ALLC}, \text{ALLC}) = (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m). \quad (\text{C. 33})$$

Because for an arbitrary opponent strategy σ , our strategy always obtains the higher or same payoff, it follows that

$$\pi(\sigma, MCSUC) \leq (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m). \quad (\text{C. 34})$$

Therefore, by [Eq. \(C. 10\)](#), we obtain the estimate

$$\begin{aligned} \pi(\sigma, MCSUC) - \pi(MCSUC, MCSUC) &\leq (1 - \epsilon)^2(R_1 + R_2 + \dots + R_m) + \\ &\epsilon(1 - \epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) + \epsilon^2(P_1 + P_2 + \dots + P_m) \\ &- \left\{ \left[\frac{2m\epsilon}{1 - 2\epsilon} + \Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right] \sum_{k=1, i \neq p_1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] + \right. \\ &\left(\Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right) [(1 - \epsilon)^2 R_{p_1} + \epsilon(1 - \epsilon)(T_{p_1} + S_{p_1}) + \epsilon^2 P_{p_1}] + \\ &\frac{m\epsilon}{1 - 2\epsilon} [2\epsilon(1 - \epsilon)(R_{p_1} + P_{p_1}) + (1 - 2\epsilon + 2\epsilon^2)(S_{p_1} + T_{p_1})] \Big\} \left(\frac{2\epsilon - 1}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1} \right) \\ &= \sum_{k=1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] \\ &- \left\{ \frac{-4\Delta\epsilon + 2\Delta + 1}{1 - 2\epsilon} \sum_{k=1, i \neq p_1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] + \right. \\ &\frac{(-1 + \epsilon)(1 + (-1 - 2m + 2m\epsilon)\epsilon + 2m\epsilon^2 + \Delta(2 - 6\epsilon + 4\epsilon^2))}{-1 + 2\epsilon} R_{p_1} + \\ &\frac{m\epsilon(-1 + 2\epsilon - 2\epsilon^2) - (-1 + \epsilon)\epsilon(-1 + 2m\epsilon + \Delta(-2 + 4\epsilon))}{-1 + 2\epsilon} (T_{p_1} + S_{p_1}) \\ &\left. \frac{\epsilon(2m\epsilon(-1 + \epsilon) + \epsilon(-1 + 2m\epsilon + \Delta(-2 + 4\epsilon)))}{-1 + 2\epsilon} P_{p_1} \right\} \left(\frac{2\epsilon - 1}{2\Delta(2\epsilon - 1) - 1} \right) \\ &= \sum_{k=1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] \\ &- \left\{ \sum_{k=1, i \neq p_1}^m [(1 - \epsilon)^2 R_k + \epsilon(1 - \epsilon)(S_k + T_k) + \epsilon^2 P_k] + \right. \\ &\frac{(-1 + \epsilon)(1 + (-1 - 2m + 2m\epsilon)\epsilon + 2m\epsilon^2 + \Delta(2 - 6\epsilon + 4\epsilon^2))}{2\Delta(2\epsilon - 1) - 1} R_{p_1} + \\ &\frac{m\epsilon(-1 + 2\epsilon - 2\epsilon^2) - (-1 + \epsilon)\epsilon(-1 + 2m\epsilon + \Delta(-2 + 4\epsilon))}{2\Delta(2\epsilon - 1) - 1} (T_{p_1} + S_{p_1}) \\ &\left. \frac{\epsilon(2m\epsilon(-1 + \epsilon) + \epsilon(-1 + 2m\epsilon + \Delta(-2 + 4\epsilon)))}{2\Delta(2\epsilon - 1) - 1} P_{p_1} \right\} \quad (\text{C. 35}) \end{aligned}$$

where p_1 here represents the channel number with the highest defection priority. Then, we can get:

$$\begin{aligned} \pi(\sigma, \text{MCSUC}) - \pi(\text{MCSUC}, \text{MCSUC}) &\leq -\frac{2m\epsilon(1-3\epsilon+2\epsilon^2)}{-1+2\Delta(-1+2\epsilon)}R_{p_1} + \frac{m(1-2\epsilon)^2\epsilon}{-1+2\Delta(-1+2\epsilon)}(T_{p_1} + S_{p_1}) \\ &+ \frac{2m(1-2\epsilon)\epsilon^2}{-1+2\Delta(-1+2\epsilon)}P_{p_1} = \frac{m\epsilon(1-2\epsilon)}{1+2\Delta(1-2\epsilon)}(2(1-\epsilon)R_{p_1} - (1-2\epsilon)(T_{p_1} + S_{p_1}) - 2\epsilon P_{p_1}) = \\ &\frac{m\epsilon(1-2\epsilon)}{1+2\Delta(1-2\epsilon)}((1-2\epsilon)(2R_{p_1} - T_{p_1} - S_{p_1}) + 2\epsilon(2R_{p_1} - P_{p_1})) \leq \frac{m}{3}\epsilon \max[(R_{p_1} - P_{p_1}), (2R_{p_1} - S_{p_1} - T_{p_1})] \leq \hat{\epsilon} \end{aligned} \quad (\text{C. 36})$$

□

C.11. Replicator dynamics among ALLD, ALLC, and MCSUC

To further explore the evolutionary characters among ALLD, ALLC, and MCSUC, we simulate the replicator dynamic analysis among these three strategies. Based on Proposition 5 and Theorems 2 and 3, the expected payoffs are captured in the following payoff matrix:

$$\pi = \begin{pmatrix} \pi_{DD} & \pi_{DC} & \pi_{DM} \\ \pi_{CD} & \pi_{CC} & \pi_{CM} \\ \pi_{MD} & \pi_{MC} & \pi_{MM} \end{pmatrix} \quad (\text{C. 37})$$

with the elements defined as follows:

$$\begin{aligned} \pi_{DD} &= \pi_{MD} = \pi_{DM} = \epsilon^2(R_1 + R_2 + \dots + R_m) + \epsilon(1-\epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) \\ &\quad + (1-\epsilon)^2(P_1 + P_2 + \dots + P_m), \\ \pi_{CC} &= \pi_{MC} = \pi_{CM} = (1-\epsilon)^2(R_1 + R_2 + \dots + R_m) + \epsilon(1-\epsilon)(T_1 + S_1 + T_2 + S_2 + \dots + T_m + S_m) \\ &\quad + \epsilon^2(P_1 + P_2 + \dots + P_m), \\ \pi_{DC} &= \epsilon(1-\epsilon)(R_1 + P_1 + R_2 + P_2 + \dots + R_m + P_m) + (1-\epsilon)^2(T_1 + T_2 + \dots + T_m) \\ &\quad + \epsilon^2(S_1 + S_2 + \dots + S_m), \\ \pi_{CD} &= \epsilon(1-\epsilon)(R_1 + P_1 + R_2 + P_2 + \dots + R_m + P_m) + \epsilon^2(T_1 + T_2 + \dots + T_m) \\ &\quad + (1-\epsilon)^2(S_1 + S_2 + \dots + S_m), \\ \pi_{MM} &= \left\{ \left[\frac{2m\epsilon}{1-2\epsilon} + \Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right] \sum_{k=1, i \neq p_1}^m [(1-\epsilon)^2 R_k + \epsilon(1-\epsilon)(S_k + T_k) + \epsilon^2 P_k] + \right. \\ &\quad \left(\Delta_{X,1} + \Delta_{Y,1} - 1 + \frac{2(m\epsilon + \epsilon - 1)}{2\epsilon - 1} \right) [(1-\epsilon)^2 R_{p_1} + \epsilon(1-\epsilon)(T_{p_1} + S_{p_1}) + \epsilon^2 P_{p_1}] + \\ &\quad \left. \frac{m\epsilon}{1-2\epsilon} \left[2\epsilon(1-\epsilon)(R_{p_1} + P_{p_1}) + (1-2\epsilon+2\epsilon^2)(S_{p_1} + T_{p_1}) \right] \right\} \left(\frac{2\epsilon - 1}{(2\epsilon - 1)(\Delta_{X,1} + \Delta_{Y,1}) - 1} \right) \end{aligned} \quad (\text{C. 38})$$

We simulate the replicator dynamics within an infinitely large population. Let x_D , x_C , and x_M represent the frequencies of ALLD, ALLC, and MCSUC, respectively. The corresponding expected payoffs for each strategy are as follows:

$$\begin{aligned} \bar{\pi}_D &= \pi_{DD} \cdot x_D + \pi_{DC} \cdot x_C + \pi_{DM} \cdot x_M \\ \bar{\pi}_C &= \pi_{CD} \cdot x_D + \pi_{CC} \cdot x_C + \pi_{CM} \cdot x_M \\ \bar{\pi}_M &= \pi_{MD} \cdot x_D + \pi_{MC} \cdot x_C + \pi_{MM} \cdot x_M \end{aligned} \quad (\text{C. 39})$$

We express the population's average payoff as $\bar{\pi} = \bar{\pi}_D x_D + \bar{\pi}_C x_C + \bar{\pi}_M x_M$. The evolutionary dynamics are described by the replicator equation:

$$\dot{x}_i = x_i(\bar{\pi}_i - \bar{\pi}) \quad (\text{C. 40})$$

In this model, the frequency of a strategy increases over time only if it performs better than the average of the population. There is a rock-paper-scissors-like dominant cycle: if MCSUC is absent, ALLD dominates; if ALLC is absent, MCSUC dominates; and if ALLD is absent, ALLC dominates. Moreover, in the limit of zero error ($\epsilon \rightarrow 0$), the equilibrium is characterized by:

$$x_D^* = 0, \quad x_C^* = \frac{R_1 - P_1 + R_2 - P_2 + \dots + R_m - P_m}{T_1 - P_1 + T_2 - P_2 + \dots + T_m - P_m}, \quad x_M^* = \frac{T_1 - R_1 + T_2 - R_2 + \dots + T_m - R_m}{T_1 - P_1 + T_2 - P_2 + \dots + T_m - P_m} \quad (\text{C. 41})$$

This implies that at equilibrium, ALLD completely vanishes, and MCSUC and ALLC coexist. For positive error rates, all such interior fixed points are encircled by closed periodic orbits, indicating that nearly all initial populations lead to oscillatory dynamics among ALLC, ALLD, and MCSUC, as shown in Fig. 4(A) and (B). Furthermore, as ϵ increases from 0.01 to 0.2 (Fig. 4(A)), the fixed point moves away from populations where ALLD nearly vanishes, suggesting that higher error rates allow for the existence of the ALLD strategy. As R increases from 2.6 to 4.0 (Fig. 4(B)), the fixed point shifts from near MCSUC to near ALLC. In other words, higher values of R result in the ALLC strategy becoming more dominant relative to MCSUC.

Additionally, when mutations are possible, the evolutionary dynamics can be modeled by the following ordinary differential equation:

$$\dot{x}_i = x_i(\bar{\pi}_i - \bar{\pi} - 3\mu) + \mu, \quad (\text{C. 42})$$

where μ represents the uniform mutation rate across all strategies. We find that the oscillations around the fixed point break down. Instead, all initial populations converge to the unique fixed point, resulting in a globally stable fixed point, as shown in Fig. 4(C). As μ increases, the fixed point moves from the edge between ALLC and MCSUC toward the center of the simplex, indicating that higher mutation rates facilitate a more heterogeneous coexistence among the strategies.

Overall, based on three-strategy population evolutionary experiments and replicator dynamics among ALLC, ALLD and MCSUC, we find that our strategy effectively suppresses the ALLD strategy and protects the ALLC strategy, thereby promoting and sustaining cooperation within evolutionary populations.

Appendix D. Numerical validation for analytical results

In the main paper, we show that if both agents adopt our proposed strategy without any “trembling hand” errors, full cooperation in all channels is guaranteed (Theorem 1). In the presence of errors, agents’ expected cooperation rate and expected payoffs are nearly optimal (Proposition 1), suggesting the resilience to the error of our proposed strategy. To validate our analytical results, we proceed to substitute numerical values into the formulas (Proposition 1, Theorems 2, and 3) for the theoretical average cooperation rate and theoretical average payoff. Additionally, we conduct simulations of interactions between two agents to obtain simulated average cooperation rates and simulated average payoffs. These numerical results confirm and validate our theoretical assertions Table D. 4.

We first determine the average cooperation rates and payoffs from the self-play of our strategy, using ($R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$), based on theoretical calculations. Considering distinct noise ϵ levels, we can calculate the average cooperation rates and payoffs for our strategy with different tolerance thresholds, shown in Table, and Table D. 1. Then we obtain the average cooperation rates and average payoffs of two agents in 1000 simulation runs, each lasting for 1,000,000 rounds, shown in Table D. 2, and Table D. 3.

The simulation data shown in Table D. 2 agree with the analytical results provided in Table 1, which numerically validate our Theorem 1 and Proposition 1. Moreover, in the main paper, we show that our proposed strategy readily retaliates and effectively avoids exploitation from the opponent when it is confronted with the ALLD (always defecting) strategy (Theorem 2). In the meantime, it is also nice and resists the temptation to defect even in the face of the ALLC (always cooperating) strategy (Theorem 3). As shown in Tables D. 1 and D. 3, when agent X using our strategy interacts with the ALLC (or ALLD) strategy agent Y , both their average cooperation rates and average payoffs are nearly the same. When Y uses ALLC, X behaves likewise; with Y as ALLD, X acts similarly. We then test the average cooperation rates and average payoffs in the multichannel IPD between an MCSUC agent and another agent who uses one of ten strategies, as shown in Table D. 5. Furthermore, we also conducted tournaments involving pairwise matches among the eleven strategies discussed in this paper and recorded the average payoffs for each pair in Table D. 6.

Table D. 1

Average cooperation rates and average payoffs obtained from self-play of our strategy with $R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$ according to theoretical calculations.

Gaming Strategy	error rate ϵ							
	No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
MCSUC ($m = 3$)	9	1	8.963605	0.988008	8.810217	0.940217	8.601429	0.880952
MCSUC ($m = 5$)	15	1	14.939341	0.988008	14.683696	0.940217	14.335714	0.880952
MCSUC ($m = 10$)	30	1	29.878683	0.988008	29.367391	0.940217	28.671429	0.880952

Table D. 2

Payoffs of our strategy with $\Delta_k = 2k$ for $k \leq m$ vs. unconditional strategies (ALLC, ALLD) with $R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$ according to theoretical calculations.

Gaming Strategy		error rate ϵ							
		No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
		Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
m = 3	MCSUC	9	1	8.9697	0.99	8.8425	0.95	8.6700	0.9
	ALLC	9	1	8.9697	0.99	8.8425	0.95	8.6700	0.9
	MCSUC	3	0	3.0897	0.01	3.4425	0.05	3.8700	0.1
	ALLD	3	0	3.0897	0.01	3.4425	0.05	3.8700	0.1
m = 5	MCSUC	15	1	14.9495	0.99	14.7375	0.95	14.4500	0.9
	ALLC	15	1	14.9495	0.99	14.7375	0.95	14.4500	0.9
	MCSUC	5	0	5.1495	0.01	5.7375	0.05	6.4500	0.1
	ALLD	5	0	5.1495	0.01	5.7375	0.05	6.4500	0.1
m = 10	MCSUC	30	1	29.8990	0.99	29.4750	0.95	28.9000	0.9
	ALLC	30	1	29.8990	0.99	29.4750	0.95	28.9000	0.9
	MCSUC	10	0	10.2990	0.01	11.4750	0.05	12.9000	0.1
	ALLD	10	0	10.2990	0.01	11.4750	0.05	12.9000	0.1

Table D. 3

Average cooperation rates and average payoffs obtained from self-play of our strategy with $R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$ according to simulations.

Gaming Strategy		error rate ϵ							
		No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
		Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
MCSUC ($m = 3$)	9	1	8.963696	0.988038	8.812516	0.940915	8.611310	0.883697	
MCSUC ($m = 5$)	15	1	14.939524	0.988043	14.688819	0.941148	14.357722	0.884621	
MCSUC ($m = 10$)	30	1	29.879277	0.988066	29.384043	0.941731	28.737506	0.886460	

Table D. 4

Payoffs of our strategy with $\Delta_k = 2k$ for $k \leq m$ vs. unconditional strategies (ALLC, ALLD) with $R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$ according to simulations.

Gaming Strategy		error rate ϵ							
		No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
		Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
m = 3	MCSUC	9	1	8.970092	0.989936	8.843328	0.949859	8.671152	0.899802
	ALLC	9	1	8.969116	0.990001	8.841207	0.950000	8.668141	0.900003
	MCSUC	2.999992	0.000002	3.089497	0.010067	3.442101	0.050142	3.869333	0.100194
	ALLD	3.000028	0	3.090505	0.010000	3.444164	0.050004	3.872282	0.099998
m = 5	MCSUC	15	1	14.950009	0.989951	14.738554	0.949893	14.451455	0.899848
	ALLC	15	1	14.948740	0.990001	14.735859	0.950001	14.447646	0.900000
	MCSUC	4.999989	0.000002	5.149263	0.010051	5.736996	0.050111	6.449271	0.100152
	ALLD	5.000044	0	5.150511	0.010001	5.739671	0.050004	6.452948	0.100005
m = 10	MCSUC	30	1	29.899725	0.989966	29.476508	0.949924	28.901898	0.899892
	ALLC	30	1	29.897945	0.990002	29.472666	0.950001	28.896734	0.899995
	MCSUC	9.999979	0.000002	10.298650	0.010038	11.474226	0.050080	12.898786	0.100106
	ALLD	10.000084	0	10.300511	0.010001	11.478128	0.050002	12.904141	0.099999

Table D. 5

Average cooperation rate and average payoffs of our strategy with $\Delta_k = 2k$ for $k \leq m$ vs. one of the ten strategies (ALLD, ALLC, HardMajority, TFT, GTFT, multichannel-WLSL, CIC, CURE, the extortion strategy, and the generous strategy) with $R_k = 3, S_k = 0, T_k = 5, P_k = 1$ for $k \leq m$ according to simulations.

Gaming Strategy		error rate ϵ							
		No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
		Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
$m = 3$	MCSUC	9	1	8.970092	0.989936	8.843328	0.949859	8.671152	0.899802
	ALLC	9	1	8.969116	0.990001	8.841207	0.950000	8.668141	0.900003
	MCSUC	2.999992	0.000002	3.089497	0.010067	3.442101	0.050142	3.869333	0.100194
	ALLD	3.000028	0	3.090505	0.010000	3.444164	0.050004	3.872282	0.099998
	MCSUC	9	1	6.222624	0.500150	6.365630	0.500053	6.500669	0.500011
	TFT	9	1	6.222647	0.500148	6.365653	0.500052	6.500691	0.500009
	MCSUC	9	1	8.896671	0.968151	8.491662	0.864860	8.094337	0.772729
	GTFT	9	1	8.896679	0.968151	8.491672	0.864860	8.094349	0.772728
	MCSUC	9	1	7.225278	0.606344	6.942296	0.544382	6.821690	0.517382
	WLSL	9	1	7.225298	0.606343	6.942317	0.544381	6.821712	0.517380
	MCSUC	9	1	5.212936	0.248111	5.199022	0.254770	5.289740	0.273052
	CIC	9	1	5.212966	0.248109	5.199053	0.254768	5.289771	0.273050
	MCSUC	8.499994	0.833333	8.069024	0.838715	7.987363	0.802627	7.916600	0.767370
	HardMajority	8.500004	0.833333	8.068497	0.838750	7.986274	0.802700	7.915254	0.767460
	MCSUC	9	1	7.773962	0.775309	7.748283	0.749029	7.698116	0.718427
	CURE	9	1	7.773974	0.775309	7.748295	0.749028	7.698128	0.718426
	MCSUC	3.000061	0.000013	3.313240	0.039748	4.255645	0.160911	5.011381	0.258620
	Extortion Strategy	3.000096	0.000011	3.313278	0.039745	4.255680	0.160908	5.011415	0.258618
	MCSUC	9	1	8.704447	0.934642	7.982241	0.774789	7.545053	0.674621
	Generous Strategy	9	1	8.704456	0.934641	7.982255	0.774788	7.545069	0.674620
$m = 5$	MCSUC	15	1	14.950009	0.989951	14.738554	0.949893	14.451455	0.899848
	ALLC	15	1	14.948740	0.990001	14.735859	0.950001	14.447646	0.900000
	MCSUC	4.999989	0.000002	5.149263	0.010051	5.736996	0.050111	6.449271	0.100152
	ALLD	5.000044	0	5.150511	0.010001	5.739671	0.050004	6.452948	0.100005
	MCSUC	15	1	10.253313	0.500129	10.522068	0.500015	10.772266	0.500022
	TFT	15	1	10.253345	0.500127	10.522100	0.500014	10.772298	0.500020
	MCSUC	15	1	14.821126	0.968154	14.116584	0.864877	13.460515	0.772728
	GTFT	15	1	14.821135	0.968154	14.116596	0.864876	13.460530	0.772727
	MCSUC	15	1	11.803635	0.574326	11.445156	0.527025	11.316166	0.509626
	WLSL	15	1	11.803664	0.574325	11.445187	0.527023	11.316198	0.509624
	MCSUC	15	1	8.001324	0.201857	8.046209	0.210568	8.154600	0.223221
	CIC	15	1	8.001372	0.201855	8.046258	0.210566	8.154651	0.223218
	MCSUC	12.999990	0.8	12.918846	0.784593	12.772513	0.744793	12.745096	0.718219
	HardMajority	13.000010	0.8	12.918857	0.784592	12.772428	0.744797	12.744697	0.718235
	MCSUC	15	1	11.544199	0.636003	11.649780	0.623848	11.762589	0.612515
	CURE	15	1	11.544224	0.636002	11.649805	0.623847	11.762614	0.612514
	MCSUC	5.000100	0.000013	5.521698	0.039834	7.088784	0.161661	8.344534	0.259217
	Extortion Strategy	5.000155	0.000011	5.521755	0.039832	7.088836	0.161659	8.344582	0.259215
	MCSUC	15	1	14.506826	0.934894	13.296899	0.775952	12.563574	0.675190
	Generous Strategy	15	1	14.506836	0.934894	13.296917	0.775952	12.563595	0.675390

Table D. 5

Continued.

Gaming Strategy		error rate ϵ							
		No noise ($\epsilon = 0\%$)		Low ($\epsilon = 1\%$)		Medium ($\epsilon = 5\%$)		High ($\epsilon = 10\%$)	
		Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates	Payoff	Cooperation rates
m = 10	MCSUC	9	1	8.970092	0.989936	8.843328	0.949859	8.671152	0.899802
	ALLC	9	1	8.969116	0.990001	8.841207	0.950000	8.668141	0.900003
	MCSUC	2.999992	0.000002	3.089497	0.010067	3.442101	0.050142	3.869333	0.100194
	ALLD	3.000028	0	3.090505	0.010000	3.444164	0.050004	3.872282	0.099998
	MCSUC	30	1	20.340086	0.500066	20.914172	0.500014	21.436873	0.500007
	TFT	30	1	20.340144	0.500065	20.914230	0.500012	21.436931	0.500005
	MCSUC	30	1	29.617279	0.968160	28.170791	0.864863	26.869889	0.772727
	GTFT	30	1	29.617289	0.968160	28.170807	0.864862	26.869912	0.772726
	MCSUC	30	1	23.165753	0.544689	22.687850	0.513006	22.563127	0.504592
	WSLS	30	1	23.165806	0.544688	22.687906	0.513004	22.563184	0.504591
	MCSUC	30	1	14.933277	0.165681	14.687640	0.161108	14.615954	0.161481
	CIC	30	1	14.933368	0.165679	14.687736	0.161106	14.616059	0.161479
	MCSUC	23.999990	0.700000	24.254886	0.705506	24.504599	0.693684	24.523476	0.671353
	HardMajority	24.000025	0.700000	24.254922	0.705505	24.504635	0.693683	24.523492	0.671352
	MCSUC	30	1	20.888068	0.531171	21.171454	0.522724	21.605145	0.521666
	CURE	30	1	20.888124	0.531169	21.171509	0.522723	21.605201	0.521665
	MCSUC	10.000194	0.000012	11.042891	0.040155	14.165291	0.162596	16.668905	0.259937
	Extortion Strategy	10.000299	0.00001	11.042996	0.040153	14.165386	0.162594	16.668992	0.259935
	MCSUC	30	1	29.012416	0.935598	26.575035	0.777422	25.100097	0.676420
	Generous Strategy	30	1	29.012428	0.935598	26.575062	0.777421	25.100132	0.676420

Table D. 6

Average payoffs in 1v1 matches between any two of the eleven strategies (MCSUC, ALLD, ALLC, HardMajority, TFT, GTFT, multichannel-WSLS, CIC, CURE, the extortion strategy, and the generous strategy), with each element representing the payoff of the column strategy when playing against the row strategy. The experiment is conducted in a 3-channel IPD with $R_k = 3$, $S_k = 0$, $T_k = 5$, $P_k = 1$ for $k \leq m$, and error rate $\epsilon = 0.1$.

	MCSUC	ALLC	ALLD	TFT	GTFT	WSLS	CIC	HardMajority	CURE	Extortion	Generous
MCSUC	–	8.970068	3.089516	6.222580	8.896650	7.225219	5.212961	7.989243	7.771844	3.313198	8.704355
ALLC	8.969133	–	0.120301	8.881188	8.907753	4.545402	0.121618	8.969702	8.968713	6.666344	8.858985
ALLD	3.090512	14.820303	–	3.206984	6.691125	8.955028	8.779151	3.089730	3.091088	3.195162	5.792601
TFT	6.222603	9.028217	3.059987	–	8.730354	6.750061	5.884462	6.044173	6.750928	3.430179	8.601361
GTFT	8.896659	9.010652	2.178072	8.280235	–	8.280246	4.347815	8.737318	8.901430	5.499359	8.773634
WSLS	7.225238	11.894742	1.604960	6.750088	8.730359	–	8.779128	1.789004	6.750100	4.995144	8.448422
CIC	5.212991	14.819425	1.649514	5.937344	7.632853	1.649533	–	1.649535	5.870350	4.595326	6.736288
HardMajority	7.988789	8.969691	3.089678	6.044183	8.819094	8.876154	8.779125	–	8.968701	4.014334	7.894996
CURE	7.771856	8.970343	3.089343	6.750890	8.901406	6.750063	5.870309	8.970375	–	3.313107	8.704184
Extortion	3.313236	10.492499	3.062963	3.561813	7.961694	6.835492	5.993812	4.816385	3.313161	–	7.030377
Generous	8.704364	9.042878	2.405515	8.479312	8.854172	8.136118	4.393646	6.956612	8.704209	5.060235	–

Appendix E. Analysis of strategy characteristics

To summarize whether the strategies mentioned in the main text satisfy niceness, retaliation, forgiveness, and invincibility (with or without the presence of errors). We first describe the definition of strategy principles for various successful strategies [9,18,19]. Then present the description of the traditional IPD strategies including always cooperating strategy (ALLC), always defecting strategy (ALLD), Tit-For-Tat (TFT) [2], Tit-For-Two-Tat (TF2T) [2], Generous Tit-For-Tat (GTFT) [7], HardMajority strategy [22], cumulative reciprocal strategy (CURE) [21], zero-determinant (ZD) strategy [8], extortion strategy as well as Generous strategy [8], invincible strategy [9], good strategy [39] and the multichannel strategies including multichannel-WSLS as well as multichannel-CIC [20], our proposed strategy MCSUC. Moreover, we summarize whether these strategies satisfy the strategy principles and provide remarks based on the strategy characteristics. Finally, we conclude that our proposed strategy MCSUC is a novel, nice, retaliatory, forgiving, and invincible strategy for the multichannel IPD.

The following are the definitions of the strategy principles in traditional IPD, following [9,18,19]:

Theorem 8 (Nice strategy). *In a traditional IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is nice if agent X is never the first to defect.*

Theorem 9 (Retaliatory strategy). *In a traditional IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is retaliatory if agent X defect immediately after an “uncalled for” defection by agent Y (or, at least, will not tolerate agent Y ’s defection infinitely), so as to induce the other agent to begin or resume cooperating.*

Theorem 10 (Forgiving strategy). *In a traditional IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is forgiving if agent X sooner or later be willing to let bygones be bygones and to resume cooperation after agent Y has defected.*

Theorem 11 (Invincible strategy). *In a traditional IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . In the presence (or absence) of errors, strategy σ_1 is invincible with the presence of errors (or without the presence of errors) if against any other strategy σ_2 , the expected payoffs r_X of agent X and r_Y of agent Y satisfy $r_X \geq r_Y$.*

We extend the above principles to multichannel IPD as follows:

Theorem 12 (Nice strategy). *In a multi-channel IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is nice if agent X never defects in any channel unless agent Y has defected first in at least one channel.*

Theorem 13 (Retaliatory strategy). *In a multi-channel IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is retaliatory if agent X defect immediately in one or more channels after an “uncalled for” defection by agent Y (or, at least, will not tolerate agent Y ’s defection infinitely), so as to induce the other agent to begin or resume cooperating.*

Theorem 14 (Forgiving strategy). *In a multi-channel IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . Strategy σ_1 is forgiving if agent X sooner or later be willing to let bygones be bygones and to resume cooperation in one or more channels after agent Y has defected.*

Theorem 15 (Invincible strategy). *In a multi-channel IPD, suppose agent X uses a strategy σ_1 while the opponent agent Y uses a strategy σ_2 . In the presence (absence) of errors, strategy σ_1 is invincible with the presence of errors (without the presence of errors) if against any other strategy σ_2 , the total expected payoffs obtained from all channels r_X of agent X and r_Y of agent Y satisfy $r_X \geq r_Y$.*

Then we summarize whether the aforementioned strategies satisfy the principles of niceness, retaliation, forgiveness, and invincibility (with or without the presence of errors) and provide remarks in the following. The symbols $\checkmark$, $\times$, and $-$ respectively represent satisfied, not satisfied, and a portion of strategies that are satisfied while others are not.

	Strategies	Description	Niceness	Retaliation	Forgiveness	Invincibility (without error)	Invincibility (with error)	Remark
Traditional IPD	ALLC	Always cooperates	✓	×	✓	×	×	Vulnerable to any strategies with defection tendencies, yet never try to retaliate.
	ALLD	Always defects	×	✓	×	✓	✓	Won't be exploited by any strategy, nor will it achieve cooperation with any strategy.
	TFT	First cooperates, then subsequently replicates the opponent's previous action	✓	✓	✓	✓	×	Vulnerable to errors, will be exploited in the presence of error.
	TF2T	First cooperates, then cooperates unless the opponent defects on the previous two moves	✓	✓	✓	×	×	More forgiving than TFT, but will be exploited by discontinuous defections.
	GTFT	A variation of TFT, allows for cooperation even after an opponent defects	✓	✓	✓	×	×	Balances forgiveness and retaliation, but will be exploited by strategies with defection tendencies.
	HardMajority	Defects unless the opponent has more cooperations than defections	×	✓	✓	×	×	Avoids exploitation at a high level but can't avoid it at a lower level.
	CURE	Cooperates unless the defection difference between the opponent and oneself exceeds one's own tolerance threshold	✓	✓	✓	✓	✓	Ensure equal payoffs with any opponent, but can't outperform any opponent.
	Zero-determinant (ZD) Strategy	Be able to unilaterally enforce a linear relationship between the players' payoffs	—	✓	—	—	×	A series of strategies, but none is invincible in the presence of error.
	Extortion Strategy	A subset of ZD strategies that is able to enforce an extortionate linear relation between agents' payoffs	×	✓	—	✓	×	Can't ensure never the first to defect, also not invincible in the presence of error.
	Generous Strategy	A subset of ZD strategies that is able to enforce a generous linear relation between agents' payoffs	✓	—	✓	×	×	Never the first to defect, but not invincible regardless in the presence of errors.
	Good Strategy	Aim to share the payoff for mutual cooperation, but are ready to fight back when being exploited	✓	✓	—	—	×	Not invincible with in the presence of errors.
Multichannel IPD	Invincible Strategy	Takes action based on both sides' last actions, ensures payoffs equal to or greater than any opponent's strategy with the same memory length	—	✓	—	✓	—	Chooses action based on previous actions, can't ensure invincible in the presence of error except for ALLD.
	Multichannel-WSLS	Cooperates in game k only if both the opponent agent and itself previously took the same action in that game	✓	✓	✓	×	×	'Nice', forgiving, retaliates selectively, but not invincible with or without errors.
	Multichannel-CIC	Fully cooperates in all channels if the agent and its opponent choose the same action in the last round, but will defect otherwise	✓	✓	✓	×	×	'Nice', forgiving, retaliates selectively, but not invincible with or without errors.
	MCSUC	Conduct a cumulative evaluation based on all past actions of the opponent and themselves in multichannel games to determine whether to fully cooperate, partially cooperate, or fully defect.	✓	✓	✓	✓	✓	Meet all principles, ensure payoff always equal to or greater than that of any opponent strategy, regardless of the presence of errors.

None of the aforementioned traditional IPD strategies or the multichannel IPD strategies meet all principles at the same time except for the CURE strategy and our proposed strategy MCSUC. However, even though both CURE and our proposed strategy MCSUC fulfill the principle of being invincible, there are significant differences between them. CURE can only ensure fairness, i.e. both CURE and any opponent strategy get the same expected payoff, which is one type of invincibility. In comparison, according to [Corollary 1](#) in the main text, our proposed strategy MCSUC will also never lose to any opponent strategy and sometimes even outperform in terms of the expected payoff.

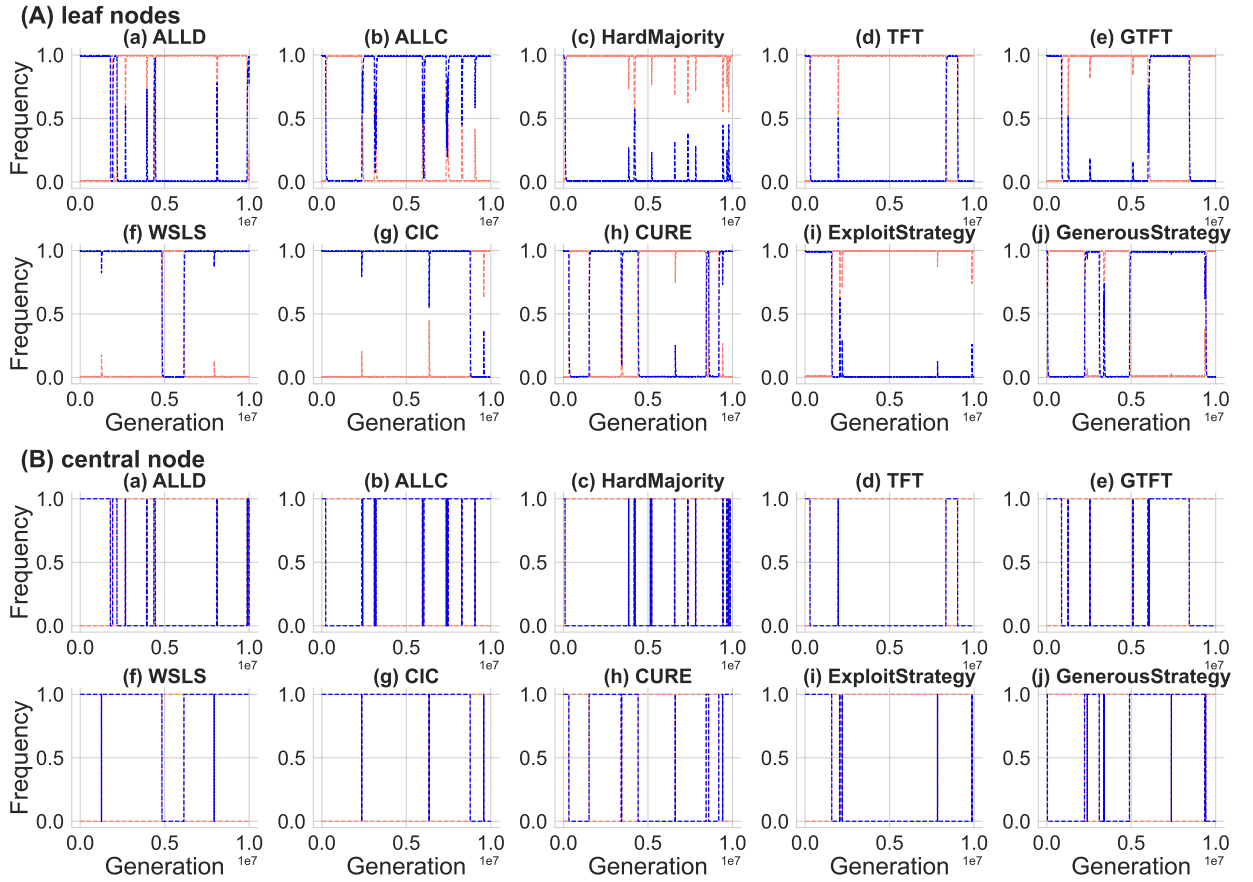


Fig. E.2. The frequency of strategy use in two-strategy populations under the star structure ((A) for leaf nodes, (B) for central node), where one strategy is MCSUC (shown in red), and the other strategy (shown in blue) is ALLD, ALLC, HardMajority, TFT, GTFT, multichannel-WSLS, CIC, CURE, the extortion strategy, or the generous strategy. The strategy frequencies of the leaf nodes are strongly influenced by the central node. As the central node frequently changes its strategy, the strategies of the leaf nodes also change over time, and the system fails to reach a stable state. Therefore, under the star structure, the evolution of the two-strategy population does not reach a stable outcome. The channel number $m = 3$. The error rate is fixed at $\epsilon = 10\%$. The mutation rate is set to 1%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

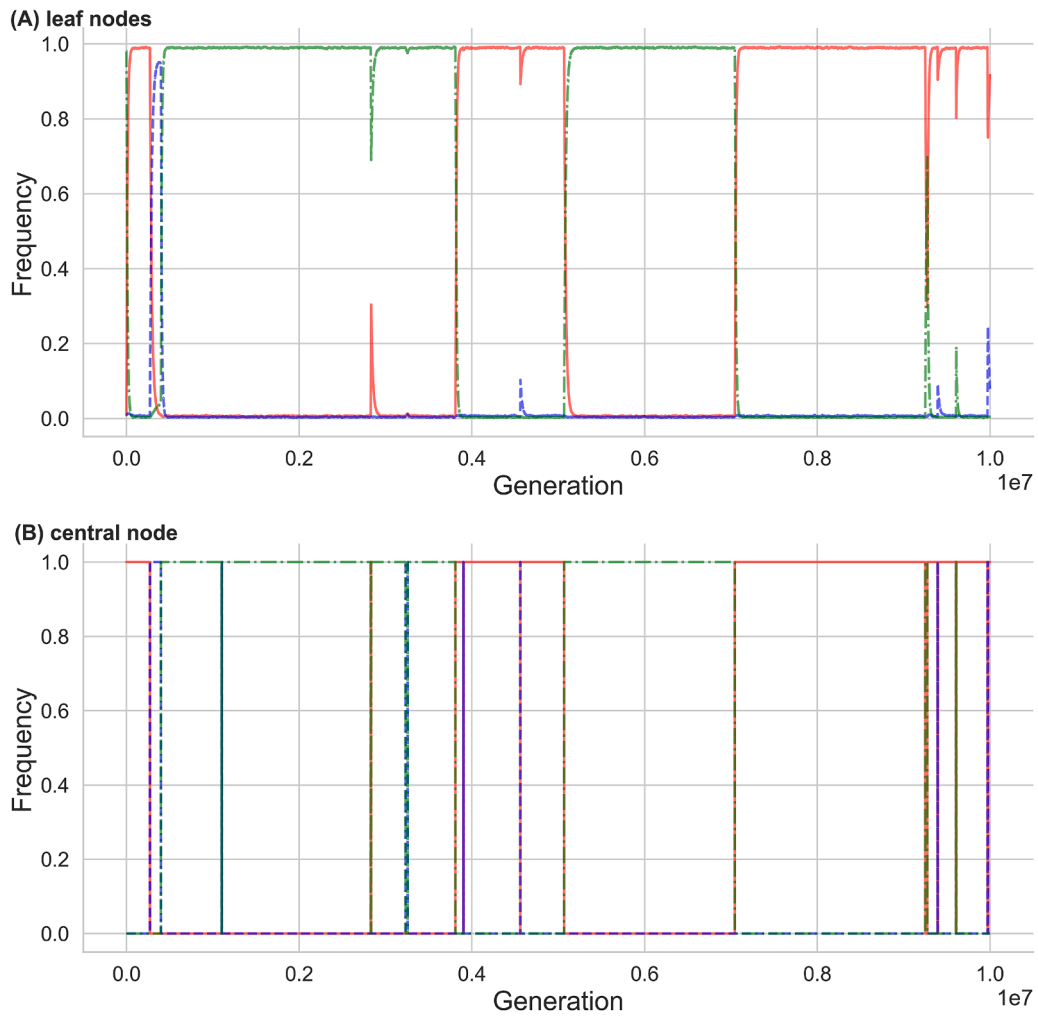


Fig. E.3. The frequency of strategy use in three-strategy populations including ALLC (shown in blue), ALLD (shown in green) and MCSUC (shown in red) with leaf nodes (A) and central node (B) under star structure. The strategy frequencies of the leaf nodes are strongly influenced by the central node. As the central node frequently changes its strategy, the strategies of the leaf nodes also change over time, and the system fails to reach a stable state. Therefore, under the star structure, the evolution of the three-strategy population does not reach a stable outcome. The channel number $m = 3$. The error rate is fixed at $\epsilon = 10\%$. The mutation rate is set to 1%. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

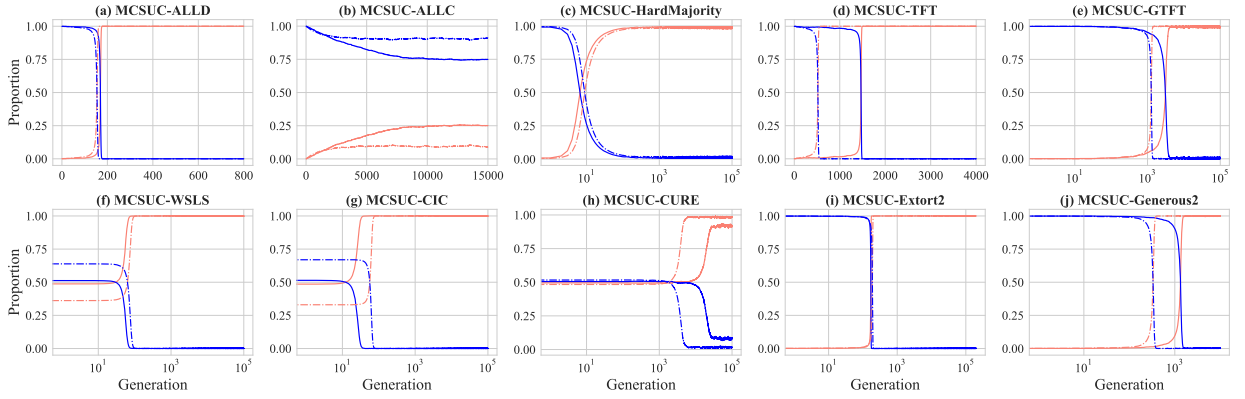


Fig. E.4. Two-strategy populations, where one strategy is MCSUC. The initial frequency of our strategy is 0.1% in most populations ((a)-(e),(i),(j)) and lower than 50% in other populations ((f),(g),(h)). There are two cases of error rates: $\epsilon = 1\%$ (solid line) and $\epsilon = 10\%$ (dashed line). The red line represents MCSUC, while the blue line represents the opponent's strategy. Our strategy is evolutionarily more advantaged than nine of the ten strategies except for the ALLC strategy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

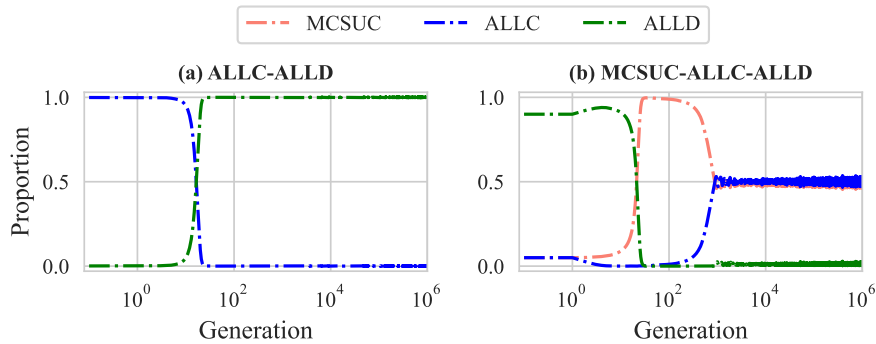


Fig. E.5. (a) A two-strategy population with only the ALLC and ALLD. The ALLD strategy invades the ALLC strategy, leading to the dominance of the ALLD strategy. (b) A three-strategy population with the ALLC, ALLD, and MCSUC. Eventually, the ALLD strategy almost disappears, and our strategy MCSUC and the ALLC strategy stably co-exist. In both subplots, the error rate is $\epsilon = 10\%$.

Appendix F. Network structure and equilibrium analysis

When the network is a star structure, where a central node connects to all other agents and each leaf node only connects to the central node, due to the heterogeneity between the central node and the leaf nodes, we no longer assume an infinite population. Instead, we consider a finite population of $n + 1$ agents, consisting of one center node and n leaf nodes. We then simulate the evolutionary dynamics between MCSUC and each of the ten other strategies (ALLD, ALLC, HardMajority, TFT, GTFT, multichannel-WSLS, CIC, CURE, the extortion strategy, and the generous strategy) in a two-strategy population, and also examine the dynamics among MCSUC, ALLC, and ALLD in a three-strategy population under the star structure.

The simulation includes two steps. First, we conduct 1000 simulation runs, each lasting 1,000,000 rounds, to estimate the long-term average payoffs of the strategies. Then, in a star network with $n = 10,000$ leaf nodes and one center node, in each simulation round, a focal agent is randomly selected. This agent can be either a leaf node or the center node. The strategy update process consists of two components. First, with a 10% probability, the focal agent undergoes a mutation and randomly switches to one of the possible strategies. Second, if no mutation occurs, the focal agent randomly selects one of its neighbors and compares average payoffs. The fitness of the focal agent and its neighbor is calculated based on the IPD with their respective neighbors. The probability that the focal agent imitates the neighbor's strategy is given by the Fermi rule:

$$P = \frac{1}{1 + \exp(-\beta(f_j - f_i))} \quad (\text{F. 1})$$

where $\beta = 1$ controls the selection intensity. The focal agent adopts the neighbor's strategy with probability P .

As shown in Fig. E.2(A) and (B), we present the results of the two-strategy population experiments. Fig. E.2(A) reports the average strategy frequency of the leaf nodes, while Fig. E.2(B) shows the strategy used by the center node. The results indicate that the population does not reach a stable state. We then conduct the three-strategy population experiment under the same star structure. Similar patterns are observed in Fig. E.3: the population fails to converge.

Appendix G. Evolutionary experiments under two-channel IPD

In this section, we present two sets of evolutionary experiments under two-channel IPD: two-strategy populations and three-strategy populations.

G.1. Two-strategy populations

To investigate if our strategy holds an evolutionary advantage, we consider two-strategy populations under two-channel IPD and let agents' strategies evolve. One strategy is MCSUC with $\Delta_1 = 2$, $\Delta_2 = 4$, and the other strategy can be multichannel-WSLS, CIC, ALLC, ALLD, TFT, GTFT, HardMajority, CURE, the extortion strategy, or the generous strategy.

For each simulation, there are two steps. First, we obtain the average payoffs of two strategies in 1000 simulation runs each lasting for 1,000,000 rounds. Then through a noisy 'survival of the fittest' environment with a mutation rate 10% [61,62], we calculate the strategies' frequencies based on the average payoffs obtained in the first step. Specifically, a noisy 'survival of the fittest' environment assumes a population of agents adopting each of the two strategies. Whether an agent changes its strategy is largely influenced by the fitness of each strategy, which is deduced from the average payoffs obtained in the first step. Our calculation of fitness follows Nowak and Sigmund's approach [7]. A strategy is said to be able to invade if the strategy starts with a low initial frequency but finally dominates the population. A simulation ends if the frequency of each strategy no longer changes. This indicates two possible steady states, that is, either the full invasion of one strategy into the other or the stable coexistence of the two strategies. It is noteworthy that different from the experiments (shown in Fig. E.4) where one agent competes against another agent, here we consider a large population of agents, in which some agents employing the MCSUC compete against other agents employing a different strategy.

As shown in Fig. E.4(a), (c)–(j), our strategy successfully invades nine out of the ten strategies, except the ALLC strategy. This means our strategy has an evolutionary advantage against these strategies except the ALLC strategy. Remarkably, even with 0.1% of agents initially adopting our strategy in the population, our strategy fully invades the ALLD, HardMajority, TFT, GTFT, extortion, and generous strategies. Moreover, with less than 50% of agents initially using our strategy, our strategy fully invades the multichannel-WSLS, CIC, and CURE strategy. We also note that our strategy is risk-dominant compared to the multichannel-WSLS, CIC, and CURE strategies.

When playing against the ALLC strategy (Fig. E.4(b)), our strategy does not invade. Rather, at the end of the simulation, our strategy and the ALLC strategy stably co-exist. We remark that failures in an invasion against the ALLC strategy are common in the research on IPD. Moreover, the ALLC strategy itself can be invaded by the ALLD strategy (Fig. E.5(a)), the latter of which, in the other way around, can be invaded by our strategy. This motivates us to ask what if the ALLC, ALLD, and MCSUC coexist?

G.2. Three-strategy population

To answer the above question, we then consider a three-strategy population (with the ALLC, ALLD, and MCSUC) using a similar approach as in the two-strategy populations. As shown in Fig. E.5(b), after 70 generations, the ALLD strategy is almost eliminated from the population, while most agents in the population adopt our strategy. Later, as our strategy cannot invade the ALLC strategy, the number of agents adopting the ALLC strategy increases, which eventually leads to the stable co-existence of our strategy and the

ALLC strategy. Because our strategy behaves like the ALLC strategy when facing the ALLC strategy, the eventual stable co-existence suggests that most of the agents will cooperate. Therefore, although our strategy cannot invade the ALLC strategy in the two-strategy population, our strategy can protect the ALLC strategy from exploitation in the three-strategy population with the presence of the ALLD strategy, thereby promoting and sustaining cooperation.

Appendix H. Examples of MCSUC applications

In this section we provide two illustrative examples to show how the multi-channel structure and cumulative reciprocity of MCSUC enable targeted responses that deter persistent opportunism while maintaining overall cooperation.

- If Microsoft and Google were to interact in the fields of search engines, cloud computing, and artificial intelligence, and if Google were to continue adopting adversarial behavior (defection) in the search engine domain, the MCSUC strategy would calculate the cumulative difference between Google's defections and Microsoft's defections. Once this difference were to exceed a preset tolerance threshold, MCSUC would guide Microsoft to retaliate selectively, prioritizing the cloud computing domain, which would have the greatest impact on their payoff. This would send a strong signal to Google while avoiding a full-scale confrontation and ensuring that Microsoft's long-term payoff would not be worse than Google's.
- If two pharmaceutical companies (X and Y) were to collaborate in data sharing, trial funding, and regulatory applications, and if company Y were to continuously default on data sharing, MCSUC would guide company X to suspend trial funding (a partial defection) as a warning. If company Y were subsequently to improve and resume cooperation, MCSUC would guide company X to fully cooperate again. This mechanism of MCSUC would allow the collaboration to withstand short-term fluctuations and foster long-term cooperation.

Appendix I. Effect of the number of channels m

The extension from single-channel to multi-channel is significant for two key reasons:

- First, many real-world systems are inherently multi-channel (we have provided two concrete examples in the response shown above). Multichannel interaction induces cross-channel spillovers: actions in one channel reshape payoffs, expectations, and learning signals in others—features a single-channel model cannot capture.
- Second, empirical results show that varying the number of channels, given by m , can result in significant differences in players' expected payoffs and cooperation rates. We ran new experiments between MCSUC and WSLs, GTFT, and TFT as examples to show the difference caused by different number of channels m with payoff matrices given by Table I. 1. Fig. I. 1 shows that increasing from $m=1$ to $m=2$ significantly changes outcomes. Total and average payoffs of MCSUC increase markedly, with Channel 1's payoff remaining almost unchanged from $m=1$ to $m=2$, while the newly added Channel 2 yields much higher payoffs but a noticeably lower cooperation rate than Channel 1 in both $m=1$ and $m=2$ settings. Similarly, Fig. I. 2 shows that changing m from 2 to 3 also alters both payoffs and cooperation. These differences are driven by the additionally introduced payoff matrix when $m = 3$ and by MCSUC's decisions based on the cumulative difference between channels.

Table I. 1

Payoff matrices for each channel in our experiments. For a given m , the game uses the first m channels in order (i.e., $m=1$ uses Channel 1 only; $m=2$ uses Channels 1–2; $m=3$ uses Channels 1–3).

Channel	R (C,C)	S (C,D)	T (D,C)	P (D,D)
Channel 1	(3, 3)	(0, 5)	(5, 0)	(1, 1)
Channel 2	(8, 8)	(0, 12)	(12, 0)	(4, 4)
Channel 3	(12, 12)	(0, 16)	(16, 0)	(4, 4)

The reason why m appeared to have little effect in our earlier experiments was because we considered a special case: all channels shared identical payoffs, rendering m largely irrelevant by design.

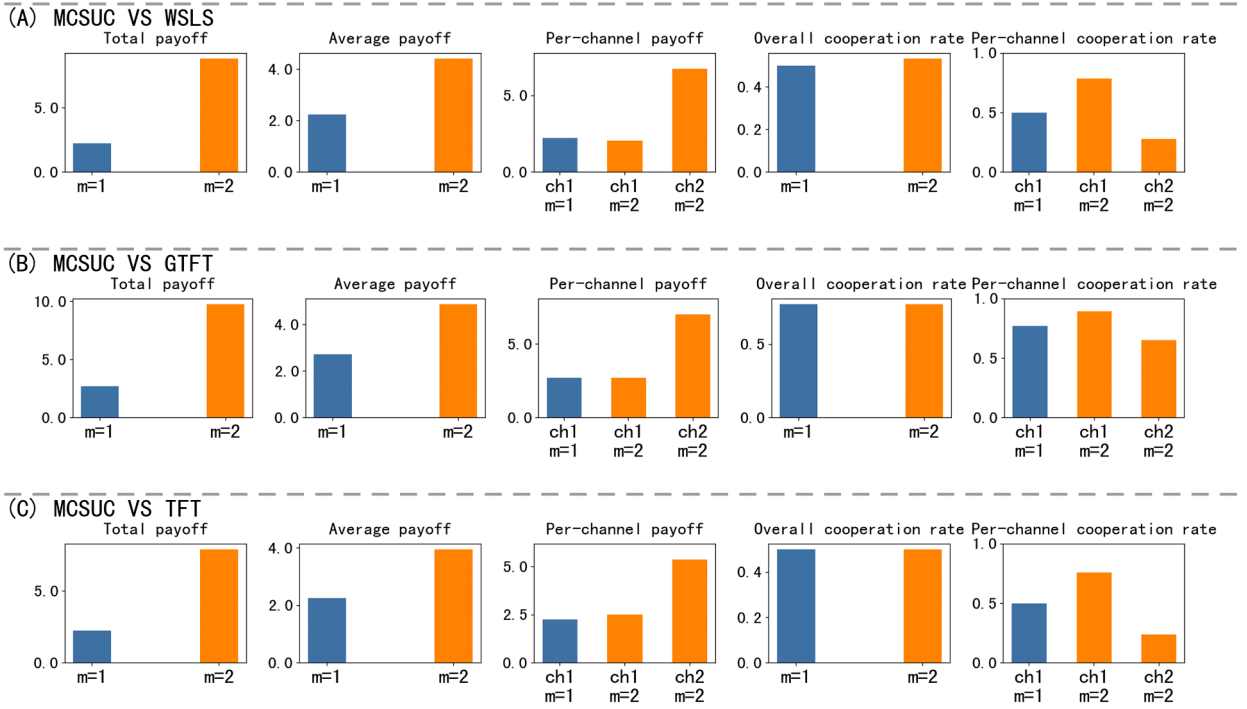


Fig. I. 1. The payoff and cooperation rates of MCSUC in multichannel ($m = 1$, $m = 2$) IPD against WSLs, GTFT, and TFT. Rows (A–C) correspond to the three opponents; bars are blue for $m = 1$ and orange for $m = 2$. From left to right the panels report: Total payoff, Average payoff, Per-channel payoff, Overall cooperation rates, and Per-channel cooperation rates. From $m = 1$ to $m = 2$, total and average payoffs increase. Channel1/s payoff remains stable, while the added Channel 2 achieves a higher payoff but lower cooperation rate than Channel 1. Overall cooperation rate is largely unchanged, though cooperation rates differ between channels. The error rate ϵ is fixed at 0.1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

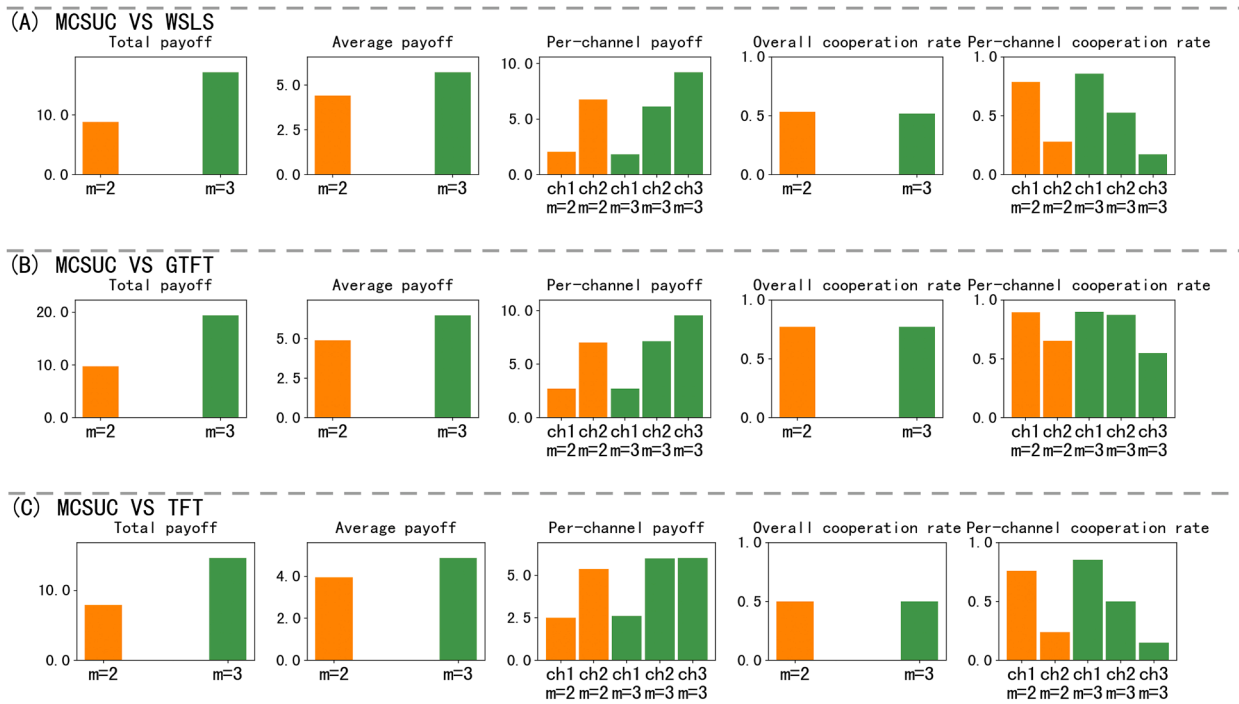


Fig. 1.2. The payoff and cooperation rates of MCSUC in multichannel ($m = 2$, $m = 3$) IPD against WLS, GTFT, and TFT. Rows (A–C) correspond to the three opponents; bars are orange for $m = 2$ and green for $m = 3$. From left to right the panels report: Total payoff, Average payoff, Per-channel payoff, Overall cooperation rates, and Per-channel cooperation rates. From $m = 2$ to $m = 3$, total and average payoffs increase. Channels 1 and 2 remain stable, whereas the added Channel 3 yields a higher payoff but lower cooperation rate than either Channel 1 or Channel 2. Overall cooperation rate remains approximately constant, though cooperation rates differ across channels. Error rate ϵ is fixed at 0.1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

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